Program Summary - Ridership_Regression_timebased.sas

Execution Environment

Author:	u64420076
File:	/home/u64420076/BAN000_Personal/Ridership_Regression_timebased.sas
SAS Platform:	Linux LIN X64 5.14.0-284.30.1.el9_2.x86_64
SAS Host:	ODAWS02-USW2.ODA.SAS.COM
SAS Version:	9.04.01M8P02222023
SAS Locale:	en_US
Submission Time:	4/16/2026, 9:27:22 PM
Browser Host:	POOL-173-35-226-51.CPE.NET.CABLE.ROGERS.COM
User Agent:	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36
Application Server:	ODAMID00-USW2.ODA.SAS.COM

Code: Ridership_Regression_timebased.sas

```
/* To remove COVID disruption I split the dataset into three distinct periods. using proc reg */


/* Importr Data */

FILENAME REFFILE '/home/u64420076/BAN000_Personal/main_dataset_modeling.csv';

PROC IMPORT DATAFILE=REFFILE
    DBMS=CSV
    OUT=WORK.TRANSIT;
    GETNAMES=YES;
RUN;

/* dummy variables */

data transit_reg;
    set transit;

    /* Month dummies: month 12 is reference */
    month_1  = (month = 1);
    month_2  = (month = 2);
    month_3  = (month = 3);
    month_4  = (month = 4);
    month_5  = (month = 5);
    month_6  = (month = 6);
    month_7  = (month = 7);
    month_8  = (month = 8);
    month_9  = (month = 9);
    month_10 = (month = 10);
    month_11 = (month = 11);

    /* COVID period dummies: Pre-COVID is reference */
    covidrec  = (covid_period = "COVID/Rec");
    postcovid = (covid_period = "Post-COVI");
run;

/* create period subsets */
data pre_covid2 ;
    set transit_reg;
    if covid_period = "Pre-COVID" then output pre_covid2;
run;

data covid_rec2;
    set transit_reg;
    if covid_period = "COVID/Rec" then output covid_rec2;
run;

data post_covid2;
    set transit_reg;
    if covid_period = "Post-COVI" then output post_covid2;
run;


/* Time baseline model */
proc reg data=transit_reg;
    model ridership =
```

```
        month_1 month_2 month_3 month_4 month_5 month_6
        month_7 month_8 month_9 month_10 month_11
        covidrec postcovid
        / vif tol;
    title "PROC REG Model 1: Baseline";
run; quit;


/* Lage Baseline Model */
proc reg data=transit_reg;
    model ridership =
        month_1 month_2 month_3 month_4 month_5 month_6
        month_7 month_8 month_9 month_10 month_11
        covidrec postcovid
        ridership_lag1 ridership_lag12
        / vif tol;
    title "PROC REG Model 2: Lag Baseline";
run; quit;

/* Full regression model */
proc reg data=transit_reg;
    model ridership =
        month_1 month_2 month_3 month_4 month_5 month_6
        month_7 month_8 month_9 month_10 month_11
        covidrec postcovid
        ridership_lag1 ridership_lag12
        gas_price_lag1 unemployment_rate_lag1
        avg_temp total_precip_mm total_snow_cm
        / vif tol;
    title "PROC REG Model 3: Full";
run; quit;

/*_____________________________________FULL-TIMELINE MODELS_____________________________________*/

/* pre-covid Time Baseline Model */
proc reg data=pre_covid2;
    model ridership =
        month_1 month_2 month_3 month_4 month_5 month_6
        month_7 month_8 month_9 month_10 month_11

        / vif tol;
    title "PROC REG Model 1: Baseline";
run; quit;

/* pre-covid Lag Baseline Model */
proc reg data=pre_covid2;
    model ridership =
        month_1 month_2 month_3 month_4 month_5 month_6
        month_7 month_8 month_9 month_10 month_11
        ridership_lag1 ridership_lag12
        / vif tol;
    title "PROC REG Model 2: Lag Baseline";
run; quit;

/* pre-covid full model */
proc reg data=pre_covid2;
    model ridership=
        month_1 month_2 month_3 month_4 month_5 month_6
        month_7 month_8 month_9 month_10 month_11
        ridership_lag1 ridership_lag12
        gas_price_lag1 unemployment_rate_lag1
        avg_temp total_precip_mm total_snow_cm
        / vif tol ;
     title 'Pre-Covid PROC REG Model 3';

run; quit;

/* step-wise */
proc reg data=pre_covid2;
    model ridership=
        month_1 month_2 month_3 month_4 month_5 month_6
        month_7 month_8 month_9 month_10 month_11
        covidrec postcovid
        ridership_lag1 ridership_lag12
        gas_price_lag1 unemployment_rate_lag1
        avg_temp total_precip_mm total_snow_cm
        / vif tol SELECTION=STEPWISE SLENTRY=0.05 SLSTAY=0.15;
     title 'Pre-Covid PROC REG Model 3';
```

```
run; quit;

/*_____________________________________PRE-COVID MODELS_____________________________________*/


/* Month-only Seasonal Baseline */
proc reg data=pre_covid2;
    model ridership =
        month_1 month_2 month_3 month_4 month_5 month_6
        month_7 month_8 month_9 month_10 month_11
        / vif tol;
    title "Pre-COVID Model P1: Month-only Seasonal Baseline";
run; quit;


/* Lag Memory */
proc reg data=pre_covid2;
    model ridership =
        ridership_lag1
        ridership_lag12
        / vif tol;
    title "Pre-COVID Model P2: Lag Memory Model";
run; quit;

/* Reduced External Model */
proc reg data=pre_covid2;
    model ridership =
        ridership_lag1
        ridership_lag12
        gas_price_lag1
        unemployment_rate_lag1
        / vif tol;
    title "Pre-COVID Model P3: Reduced External Model";
run; quit;


proc reg data=pre_covid2;
    model ridership=
        gas_price_lag1 unemployment_rate_lag1
        avg_temp total_precip_mm total_snow_cm
        / vif tol ;
     title 'Pre-Covid Model P4: Whithout Ridership History';

run; quit;
```

Log: Ridership_Regression_timebased.sas

Notes (27)

```
1          OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
NOTE: ODS statements in the SAS Studio environment may disable some output features.
69
70
71         /* To remove COVID disruption I split the dataset into three distinct periods. using proc reg */
72
73
74
75         /* Importr Data */
76
77         FILENAME REFFILE '/home/u64420076/BAN000_Personal/main_dataset_modeling.csv';
78
79         PROC IMPORT DATAFILE=REFFILE
80         DBMS=CSV
81         OUT=WORK.TRANSIT;
82         GETNAMES=YES;
83         RUN;

NOTE: Import cancelled.  Output dataset WORK.TRANSIT already exists.  Specify REPLACE option to overwrite it.
NOTE: The SAS System stopped processing this step because of errors.
NOTE: PROCEDURE IMPORT used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              194.59k
      OS Memory           27560.00k
      Timestamp           04/17/2026 01:27:18 AM
      Step Count                        57  Switch Count  0
      Page Faults                       0
      Page Reclaims                     14
      Page Swaps                        0
      Voluntary Context Switches        0
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           8

84
85         /* dummy variables */
86


87         data transit_reg;
88             set transit;
89
90             /* Month dummies: month 12 is reference */
91             month_1  = (month = 1);
92             month_2  = (month = 2);
93             month_3  = (month = 3);
94             month_4  = (month = 4);
95             month_5  = (month = 5);
96             month_6  = (month = 6);
97             month_7  = (month = 7);
98             month_8  = (month = 8);
99             month_9  = (month = 9);
100            month_10 = (month = 10);
101            month_11 = (month = 11);
102
103            /* COVID period dummies: Pre-COVID is reference */
104            covidrec  = (covid_period = "COVID/Rec");
105            postcovid = (covid_period = "Post-COVI");
106        run;

NOTE: There were 179 observations read from the data set WORK.TRANSIT.
NOTE: The data set WORK.TRANSIT_REG has 179 observations and 43 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              1063.90k
      OS Memory           28080.00k
      Timestamp           04/17/2026 01:27:18 AM
      Step Count                        58  Switch Count  2
      Page Faults                       0
      Page Reclaims                     105
      Page Swaps                        0
      Voluntary Context Switches        11
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           264


107
108        /* create period subsets */
109        data pre_covid2 ;
110            set transit_reg;
111            if covid_period = "Pre-COVID" then output pre_covid2;
112        run;
```

```
NOTE: There were 179 observations read from the data set WORK.TRANSIT_REG.
NOTE: The data set WORK.PRE_COVID2 has 109 observations and 43 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              1059.59k
      OS Memory           28080.00k
      Timestamp           04/17/2026 01:27:18 AM
      Step Count                        59  Switch Count  2
      Page Faults                       0
      Page Reclaims                     112
      Page Swaps                        0
      Voluntary Context Switches        11
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           272


113
114        data covid_rec2;
115            set transit_reg;
116            if covid_period = "COVID/Rec" then output covid_rec2;
117        run;

NOTE: There were 179 observations read from the data set WORK.TRANSIT_REG.
NOTE: The data set WORK.COVID_REC2 has 34 observations and 43 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.01 seconds
      memory              1058.28k
      OS Memory           28080.00k
      Timestamp           04/17/2026 01:27:18 AM
      Step Count                        60  Switch Count  2
      Page Faults                       0
      Page Reclaims                     117
      Page Swaps                        0
      Voluntary Context Switches        11
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           264


118
119        data post_covid2;
120            set transit_reg;
121            if covid_period = "Post-COVI" then output post_covid2;
122        run;

NOTE: There were 179 observations read from the data set WORK.TRANSIT_REG.
NOTE: The data set WORK.POST_COVID2 has 36 observations and 43 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              1058.28k
      OS Memory           28080.00k
      Timestamp           04/17/2026 01:27:18 AM
      Step Count                        61  Switch Count  2
      Page Faults                       0
      Page Reclaims                     117
      Page Swaps                        0
      Voluntary Context Switches        12
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           264


123
124
125        /* Time baseline model */
126        proc reg data=transit_reg;
127            model ridership =
128                month_1 month_2 month_3 month_4 month_5 month_6
129                month_7 month_8 month_9 month_10 month_11
130                covidrec postcovid
131                / vif tol;
132            title "PROC REG Model 1: Baseline";
133        run;

133      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.46 seconds
      user cpu time       0.19 seconds
      system cpu time     0.03 seconds
      memory              14591.46k
      OS Memory           38220.00k
```

```
Timestamp           04/17/2026 01:27:19 AM
Step Count                        62  Switch Count  23
Page Faults                       0
Page Reclaims                     14474
Page Swaps                        0
Voluntary Context Switches        1076
Involuntary Context Switches      8
Block Input Operations            0
Block Output Operations           1832


134
135
136        /* Lage Baseline Model */
137        proc reg data=transit_reg;
138            model ridership =
139                month_1 month_2 month_3 month_4 month_5 month_6
140                month_7 month_8 month_9 month_10 month_11
141                covidrec postcovid
142                ridership_lag1 ridership_lag12
143                / vif tol;
144            title "PROC REG Model 2: Lag Baseline";
145        run;

145      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.42 seconds
      user cpu time       0.18 seconds
      system cpu time     0.03 seconds
      memory              11705.78k
      OS Memory           43496.00k
      Timestamp           04/17/2026 01:27:19 AM
      Step Count                        63  Switch Count  23
      Page Faults                       0
      Page Reclaims                     12686
      Page Swaps                        0
      Voluntary Context Switches        1070
      Involuntary Context Switches      9
      Block Input Operations            0
      Block Output Operations           1888


146
147        /* Full regression model */
148        proc reg data=transit_reg;
149            model ridership =
150                month_1 month_2 month_3 month_4 month_5 month_6
151                month_7 month_8 month_9 month_10 month_11
152                covidrec postcovid
153                ridership_lag1 ridership_lag12
154                gas_price_lag1 unemployment_rate_lag1
155                avg_temp total_precip_mm total_snow_cm
156                / vif tol;
157            title "PROC REG Model 3: Full";
158        run;

158      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.49 seconds
      user cpu time       0.21 seconds
      system cpu time     0.03 seconds
      memory              11568.09k
      OS Memory           43496.00k
      Timestamp           04/17/2026 01:27:19 AM
      Step Count                        64  Switch Count  23
      Page Faults                       0
      Page Reclaims                     12789
      Page Swaps                        0
      Voluntary Context Switches        1199
      Involuntary Context Switches      14
      Block Input Operations            0
      Block Output Operations           2008


159
160        /*_________________________________________FULL-TIMELINE MODELS_________________________________________*/
161
162        /* pre-covid Time Baseline Model */
163        proc reg data=pre_covid2;
164            model ridership =
165                month_1 month_2 month_3 month_4 month_5 month_6
166                month_7 month_8 month_9 month_10 month_11
167
168                / vif tol;
169            title "PROC REG Model 1: Baseline";
170        run;
```

```
170      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.32 seconds
      user cpu time       0.15 seconds
      system cpu time     0.02 seconds
      memory              11371.00k
      OS Memory           43496.00k
      Timestamp           04/17/2026 01:27:20 AM
      Step Count                        65  Switch Count  23
      Page Faults                       0
      Page Reclaims                     12310
      Page Swaps                        0
      Voluntary Context Switches        884
      Involuntary Context Switches      9
      Block Input Operations            0
      Block Output Operations           992


171
172        /* pre-covid Lag Baseline Model */
173        proc reg data=pre_covid2;
174            model ridership =
175                month_1 month_2 month_3 month_4 month_5 month_6
176                month_7 month_8 month_9 month_10 month_11
177                ridership_lag1 ridership_lag12
178                / vif tol;
179            title "PROC REG Model 2: Lag Baseline";
180        run;

180      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.37 seconds
      user cpu time       0.17 seconds
      system cpu time     0.04 seconds
      memory              11279.40k
      OS Memory           43240.00k
      Timestamp           04/17/2026 01:27:20 AM
      Step Count                        66  Switch Count  23
      Page Faults                       0
      Page Reclaims                     12682
      Page Swaps                        0
      Voluntary Context Switches        1008
      Involuntary Context Switches      10
      Block Input Operations            0
      Block Output Operations           1656


181
182        /* pre-covid full model */
183        proc reg data=pre_covid2;
184        model ridership=
185        month_1 month_2 month_3 month_4 month_5 month_6
186                month_7 month_8 month_9 month_10 month_11
187                ridership_lag1 ridership_lag12
188                gas_price_lag1 unemployment_rate_lag1
189                avg_temp total_precip_mm total_snow_cm
190                / vif tol ;
191            title 'Pre-Covid PROC REG Model 3';
192
193        run;

193      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.38 seconds
      user cpu time       0.18 seconds
      system cpu time     0.03 seconds
      memory              11366.37k
      OS Memory           44008.00k
      Timestamp           04/17/2026 01:27:21 AM
      Step Count                        67  Switch Count  23
      Page Faults                       0
      Page Reclaims                     12504
      Page Swaps                        0
      Voluntary Context Switches        1068
      Involuntary Context Switches      13
      Block Input Operations            0
      Block Output Operations           1760


194
195        /* step-wise */
196        proc reg data=pre_covid2;
197        model ridership=
198        month_1 month_2 month_3 month_4 month_5 month_6
199                month_7 month_8 month_9 month_10 month_11
200                covidrec postcovid
```

```
201                ridership_lag1 ridership_lag12
202                gas_price_lag1 unemployment_rate_lag1
203                avg_temp total_precip_mm total_snow_cm
204                / vif tol SELECTION=STEPWISE SLENTRY=0.05 SLSTAY=0.15;
205             title 'Pre-Covid PROC REG Model 3';
206
207        run;

207      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.39 seconds
      user cpu time       0.26 seconds
      system cpu time     0.02 seconds
      memory              11479.31k
      OS Memory           43752.00k
      Timestamp           04/17/2026 01:27:21 AM
      Step Count                        68  Switch Count  23
      Page Faults                       0
      Page Reclaims                     12109
      Page Swaps                        0
      Voluntary Context Switches        739
      Involuntary Context Switches      5
      Block Input Operations            0
      Block Output Operations           992


208
209        /*_____________________________________PRE-COVID MODELS_____________________________________*/
210
211
212        /* Month-only Seasonal Baseline */
213        proc reg data=pre_covid2;
214            model ridership =
215                month_1 month_2 month_3 month_4 month_5 month_6
216                month_7 month_8 month_9 month_10 month_11
217                / vif tol;
218            title "Pre-COVID Model P1: Month-only Seasonal Baseline";
219        run;

219      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.30 seconds
      user cpu time       0.15 seconds
      system cpu time     0.02 seconds
      memory              11149.87k
      OS Memory           43496.00k
      Timestamp           04/17/2026 01:27:21 AM
      Step Count                        69  Switch Count  23
      Page Faults                       0
      Page Reclaims                     12260
      Page Swaps                        0
      Voluntary Context Switches        876
      Involuntary Context Switches      8
      Block Input Operations            0
      Block Output Operations           992


220
221
222        /* Lag Memory */
223        proc reg data=pre_covid2;
224            model ridership =
225                ridership_lag1
226                ridership_lag12
227                / vif tol;
228            title "Pre-COVID Model P2: Lag Memory Model";
229        run;

229      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.23 seconds
      user cpu time       0.11 seconds
      system cpu time     0.03 seconds
      memory              11354.28k
      OS Memory           43496.00k
      Timestamp           04/17/2026 01:27:21 AM
      Step Count                        70  Switch Count  23
      Page Faults                       0
      Page Reclaims                     12110
      Page Swaps                        0
      Voluntary Context Switches        694
      Involuntary Context Switches      5
      Block Input Operations            0
      Block Output Operations           896
```

```
230
231        /* Reduced External Model */
232        proc reg data=pre_covid2;
233            model ridership =
234                ridership_lag1
235                ridership_lag12
236                gas_price_lag1
237                unemployment_rate_lag1
238                / vif tol;
239            title "Pre-COVID Model P3: Reduced External Model";
240        run;

240      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.24 seconds
      user cpu time       0.13 seconds
      system cpu time     0.02 seconds
      memory              11362.96k
      OS Memory           43496.00k
      Timestamp           04/17/2026 01:27:22 AM
      Step Count                        71  Switch Count  23
      Page Faults                       0
      Page Reclaims                     12084
      Page Swaps                        0
      Voluntary Context Switches        712
      Involuntary Context Switches      5
      Block Input Operations            0
      Block Output Operations           936


241
242
243
244        proc reg data=pre_covid2;
245        model ridership=
246                gas_price_lag1 unemployment_rate_lag1
247                avg_temp total_precip_mm total_snow_cm
248                / vif tol ;
249             title 'Pre-Covid Model P4: Whithout Ridership History';
250
251        run;

251      !      quit;

NOTE: PROCEDURE REG used (Total process time):
      real time           0.25 seconds
      user cpu time       0.13 seconds
      system cpu time     0.02 seconds
      memory              11270.12k
      OS Memory           43752.00k
      Timestamp           04/17/2026 01:27:22 AM
      Step Count                        72  Switch Count  23
      Page Faults                       0
      Page Reclaims                     12041
      Page Swaps                        0
      Voluntary Context Switches        712
      Involuntary Context Switches      5
      Block Input Operations            0
      Block Output Operations           960


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```

```
279
280
281
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283
284
285        OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
295
```

Results: Ridership_Regression_timebased.sas

PROC REG Model 1: Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	179
Number of Observations Used	179

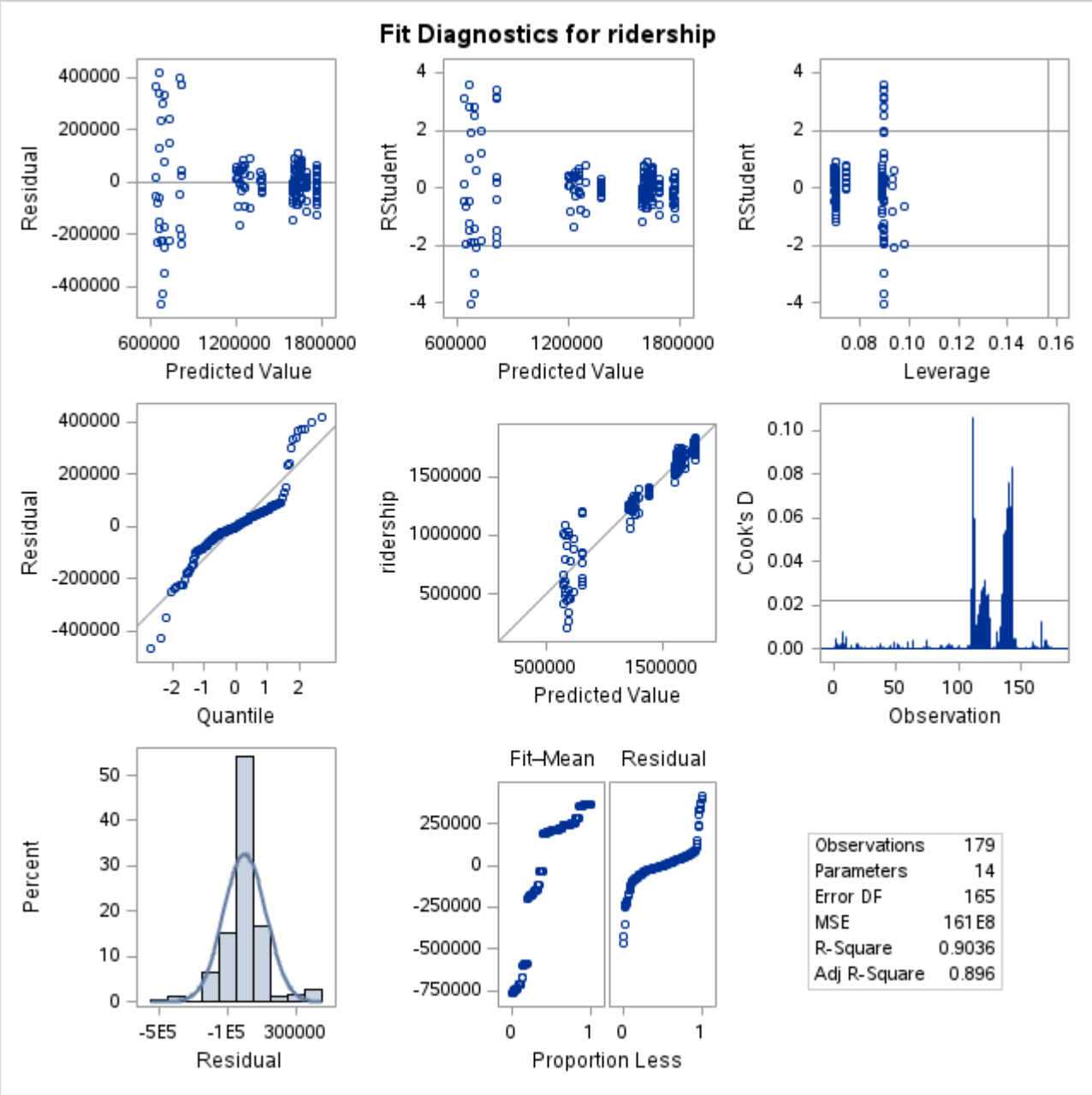
Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	13	2.489349E13	1.914884E12	118.97	<.0001
Error	165	2.655681E12	16095037667		
Corrected Total	178	2.754917E13			

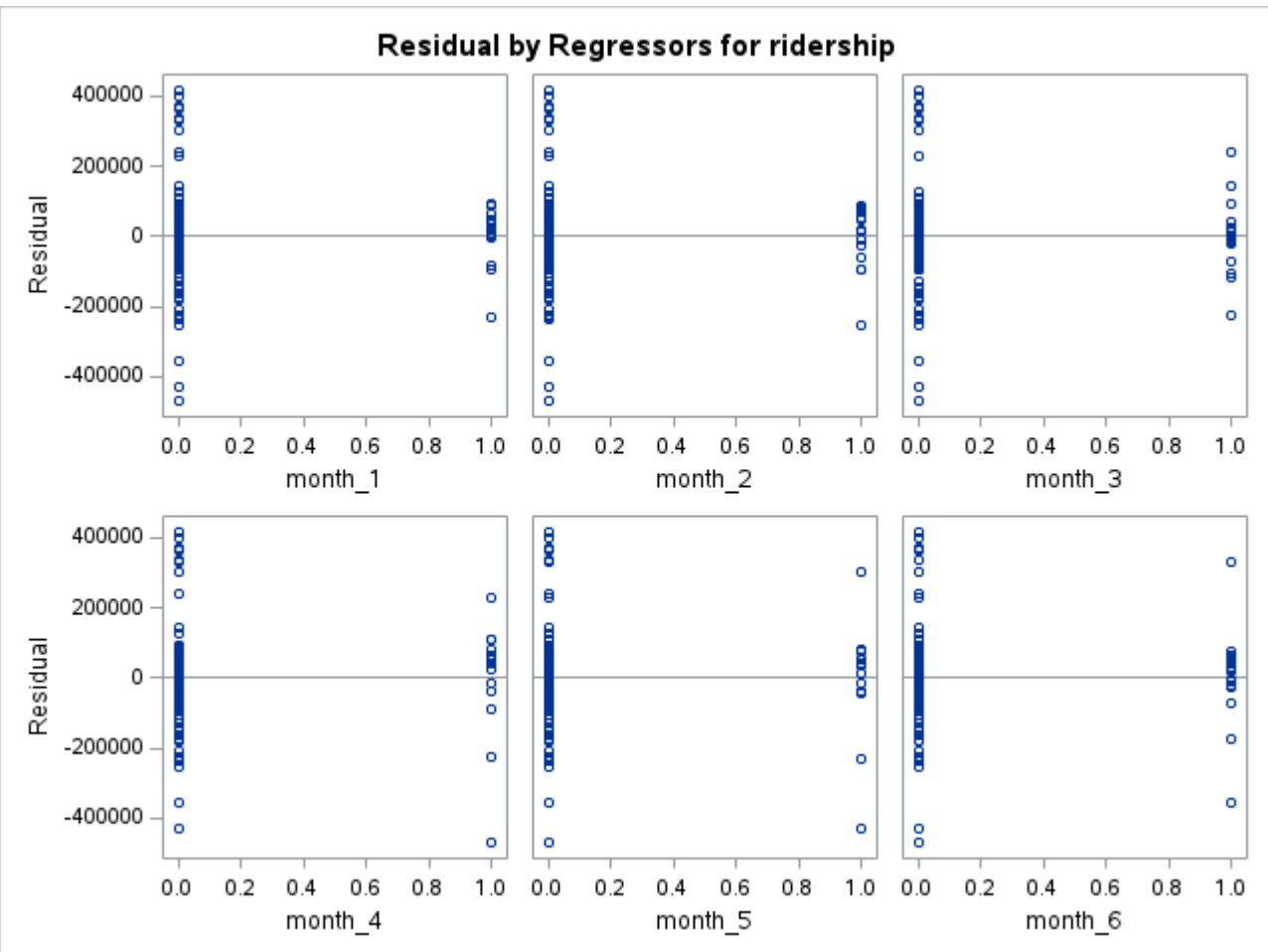
Root MSE	126866	R-Square	0.9036
Dependent Mean	1406726	Adj R-Sq	0.8960
Coeff Var	9.01854		

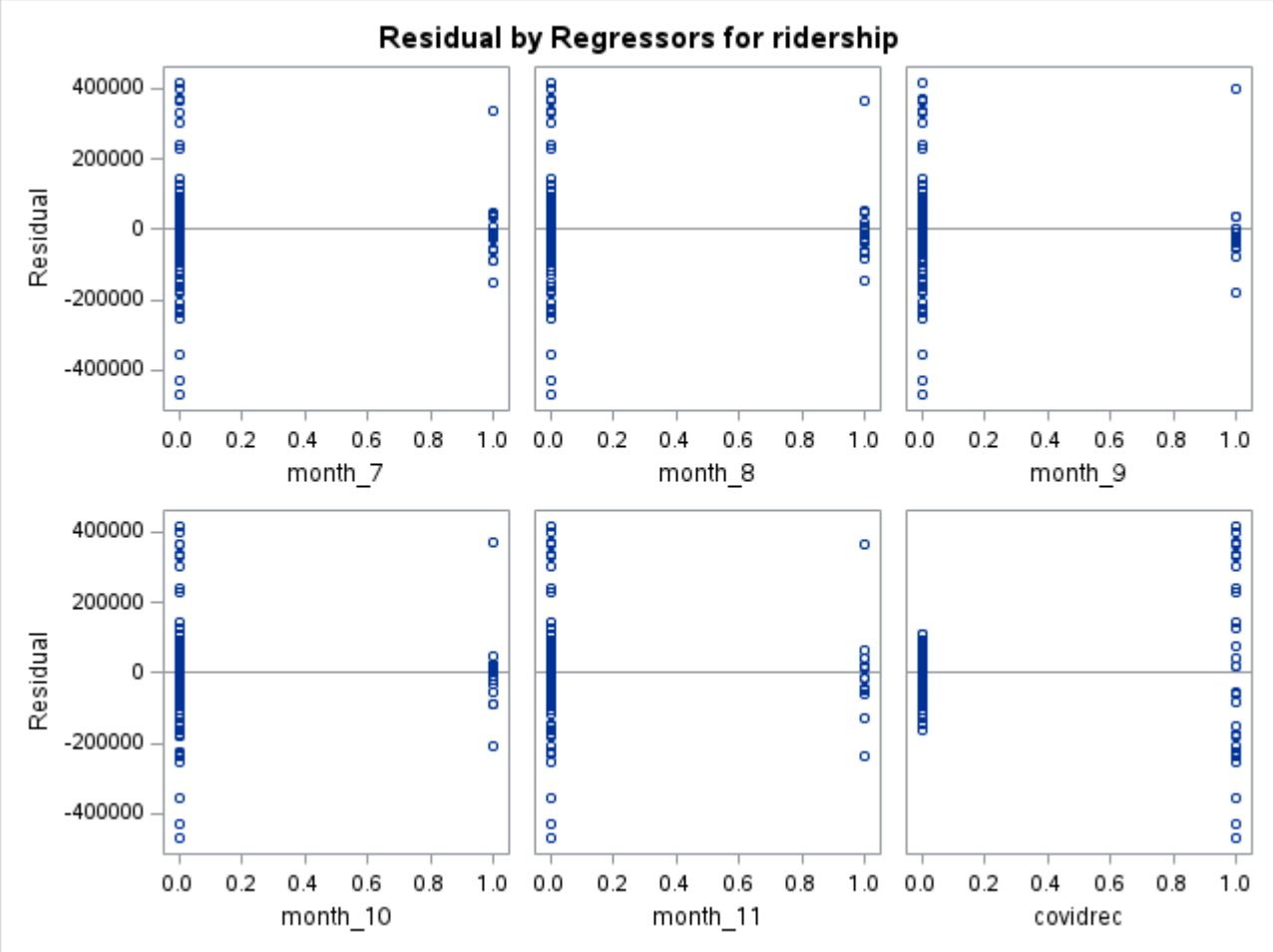
Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	1616007	33668	48.00	<.0001	.	0
month_1	1	-11129	47165	-0.24	0.8138	0.56065	1.78366
month_2	1	37679	46355	0.81	0.4175	0.54503	1.83477
month_3	1	69267	46325	1.50	0.1368	0.54573	1.83240
month_4	1	11133	46325	0.24	0.8104	0.54573	1.83240
month_5	1	28800	46325	0.62	0.5350	0.54573	1.83240
month_6	1	34333	46325	0.74	0.4597	0.54573	1.83240
month_7	1	1133.33333	46325	0.02	0.9805	0.54573	1.83240
month_8	1	-20133	46325	-0.43	0.6644	0.54573	1.83240
month_9	1	147400	46325	3.18	0.0017	0.54573	1.83240
month_10	1	152000	46325	3.28	0.0013	0.54573	1.83240
month_11	1	150800	46325	3.26	0.0014	0.54573	1.83240
covidrec	1	-952810	24964	-38.17	<.0001	0.93770	1.06644
postcovid	1	-391558	24389	-16.05	<.0001	0.94084	1.06288

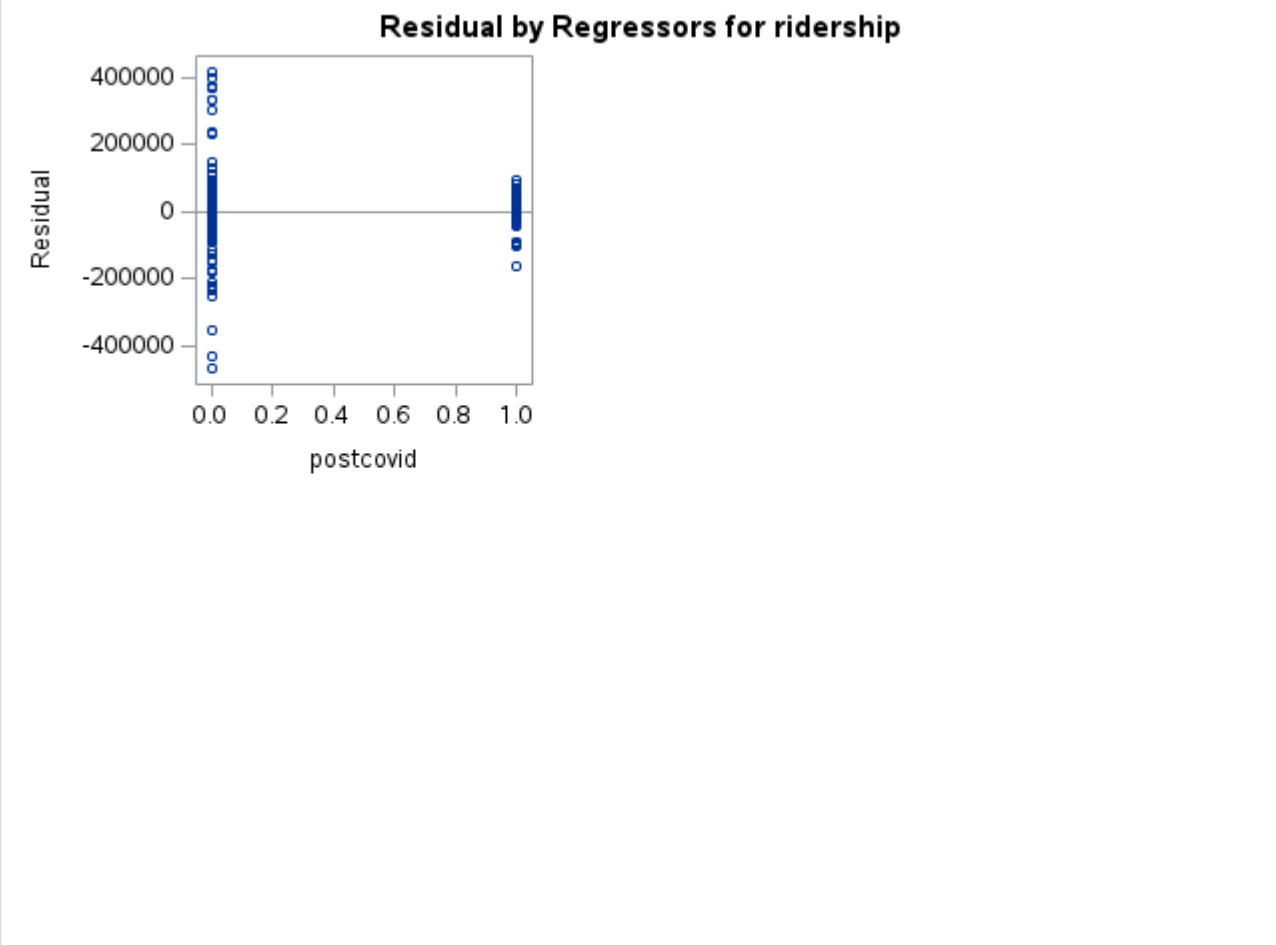
PROC REG Model 1: Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership


Fit Diagnostics for ridership
Residual
Predicted Value
RStudent
Leverage
Quantile
ridership
Cook's D
Observation
Percent
Fit–Mean
Residual
Proportion Less
Observations 179
Parameters 14
Error DF 165
MSE 161E8
R-Square 0.9036
Adj R-Square 0.896


Residual by Regressors for ridership
Residual
month_1
month_2
month_3
month_4
month_5
month_6





PROC REG Model 2: Lag Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	179
Number of Observations Used	179

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	15	2.650508E13	1.767005E12	275.86	<.0001
Error	163	1.044092E12	6405470952		
Corrected Total	178	2.754917E13			

Root MSE	80034	**R-Square**	0.9621
Dependent Mean	1406726	**Adj R-Sq**	0.9586
Coeff Var	5.68939		

Parameter Estimates

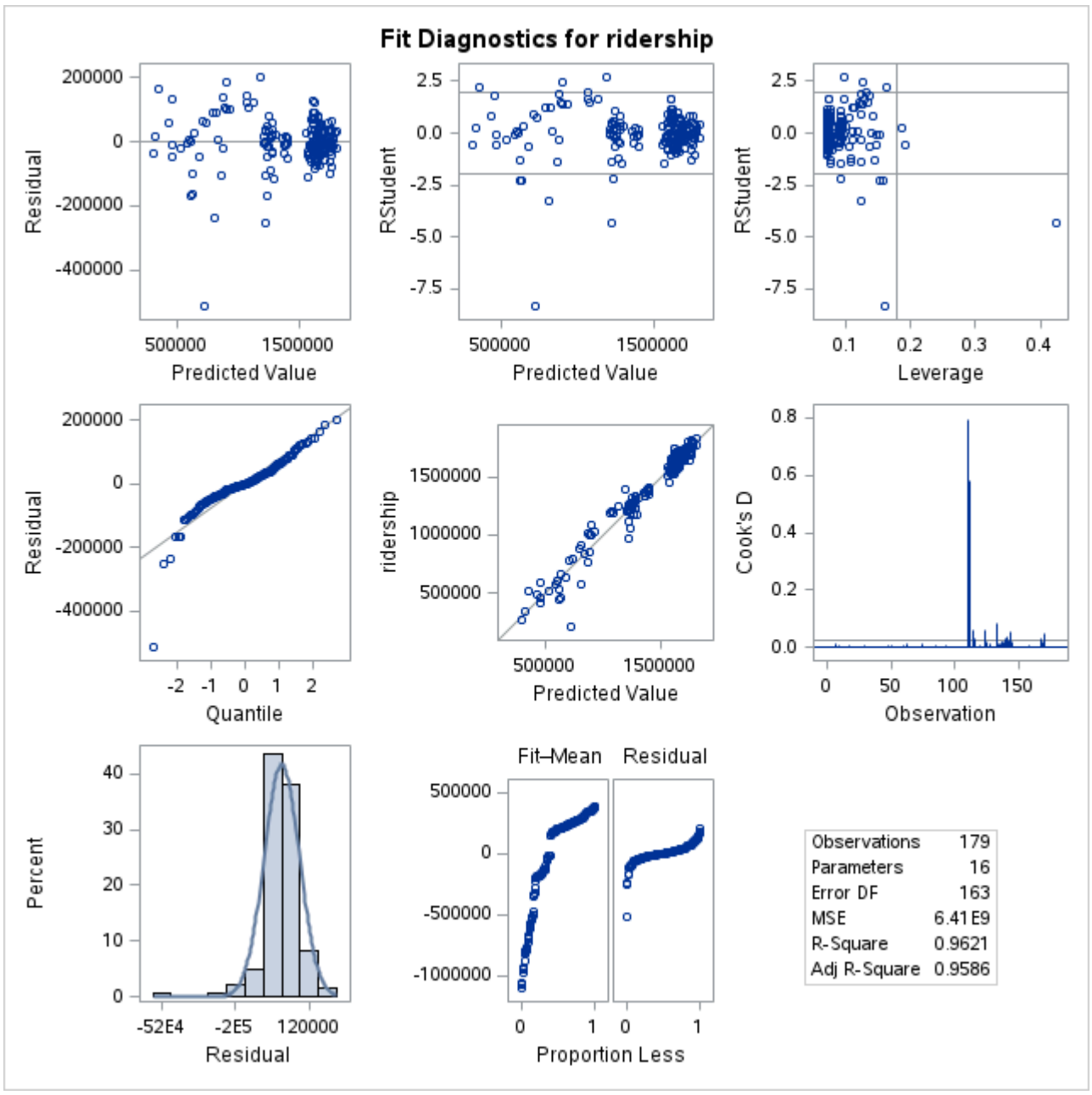
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	725924	91067	7.97	<.0001	.	0
month_1	1	106990	30859	3.47	0.0007	0.52122	1.91857
month_2	1	147419	30101	4.90	<.0001	0.51441	1.94399
month_3	1	107302	29325	3.66	0.0003	0.54200	1.84503

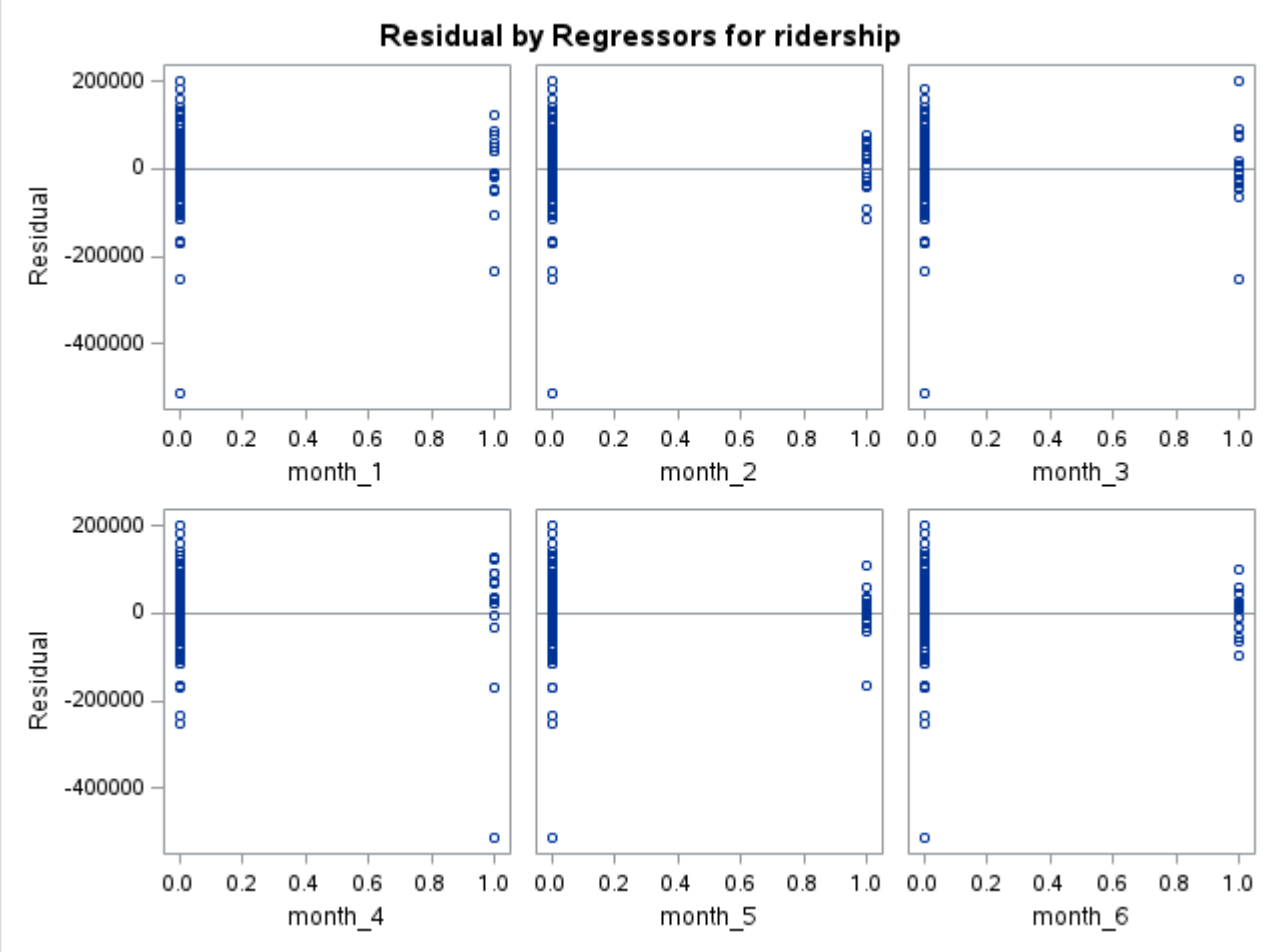
Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
month_4	1	62463	29435	2.12	0.0353	0.53795	1.85889
month_5	1	118316	29836	3.97	0.0001	0.52359	1.90988
month_6	1	113488	29692	3.82	0.0002	0.52866	1.89156
month_7	1	72724	29657	2.45	0.0153	0.52992	1.88708
month_8	1	69003	29955	2.30	0.0225	0.51945	1.92512
month_9	1	270994	30245	8.96	<.0001	0.50951	1.96267
month_10	1	173114	29439	5.88	<.0001	0.53779	1.85946
month_11	1	168247	29423	5.72	<.0001	0.53838	1.85743
covidrec	1	-460413	47755	-9.64	<.0001	0.10198	9.80567
postcovid	1	-207233	26305	-7.88	<.0001	0.32188	3.10673
ridership_lag1	1	0.61632	0.04331	14.23	<.0001	0.12404	8.06171
ridership_lag12	1	-0.12361	0.02317	-5.34	<.0001	0.43396	2.30434

PROC REG Model 2: Lag Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership




Residual by Regressors for ridership
Residual
200000
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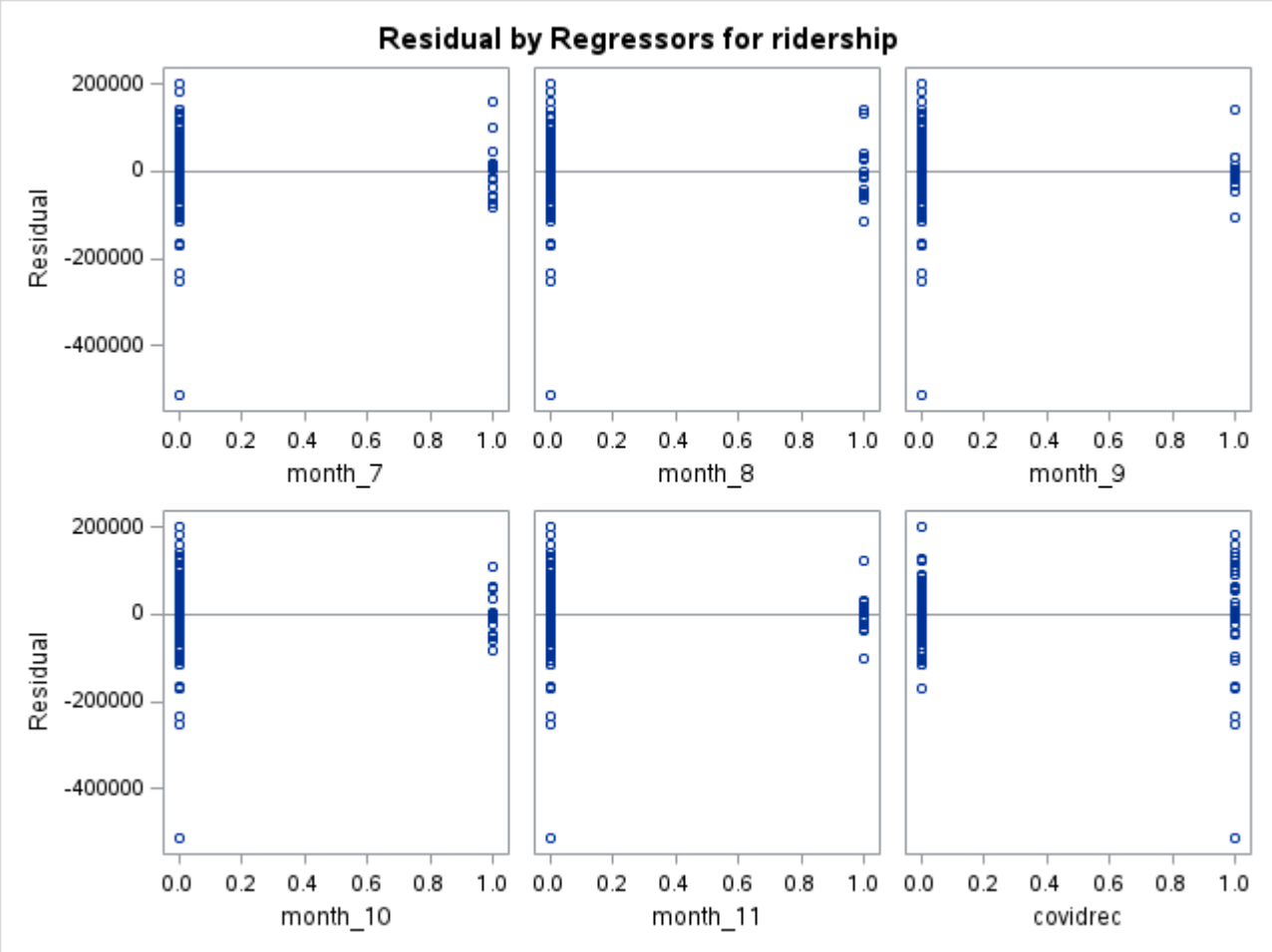
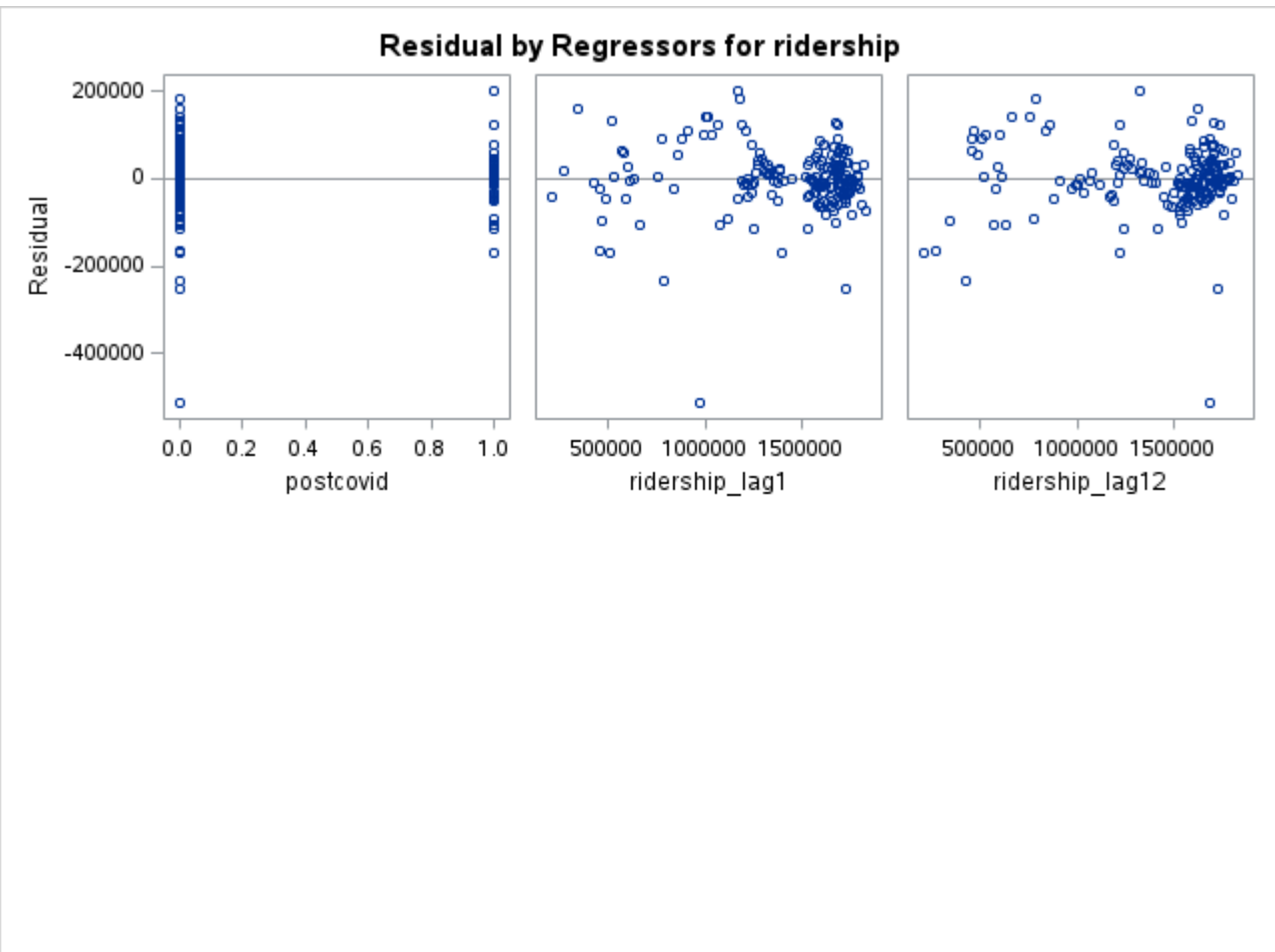
-200000
-400000
0.0 0.2 0.4 0.6 0.8 1.0
month_1
month_2
month_3
month_4
month_5
month_6


Residual by Regressors for ridership
Residual
200000
0
-200000
-400000
0.0 0.2 0.4 0.6 0.8 1.0
month_7
month_8
month_9
month_10
month_11
covidrec


Residual by Regressors for ridership
Residual
200000
0
-200000
-400000
0.0 0.2 0.4 0.6 0.8 1.0
postcovid
500000 1000000 1500000
ridership_lag1
ridership_lag12

PROC REG Model 3: Full

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	179
Number of Observations Used	179

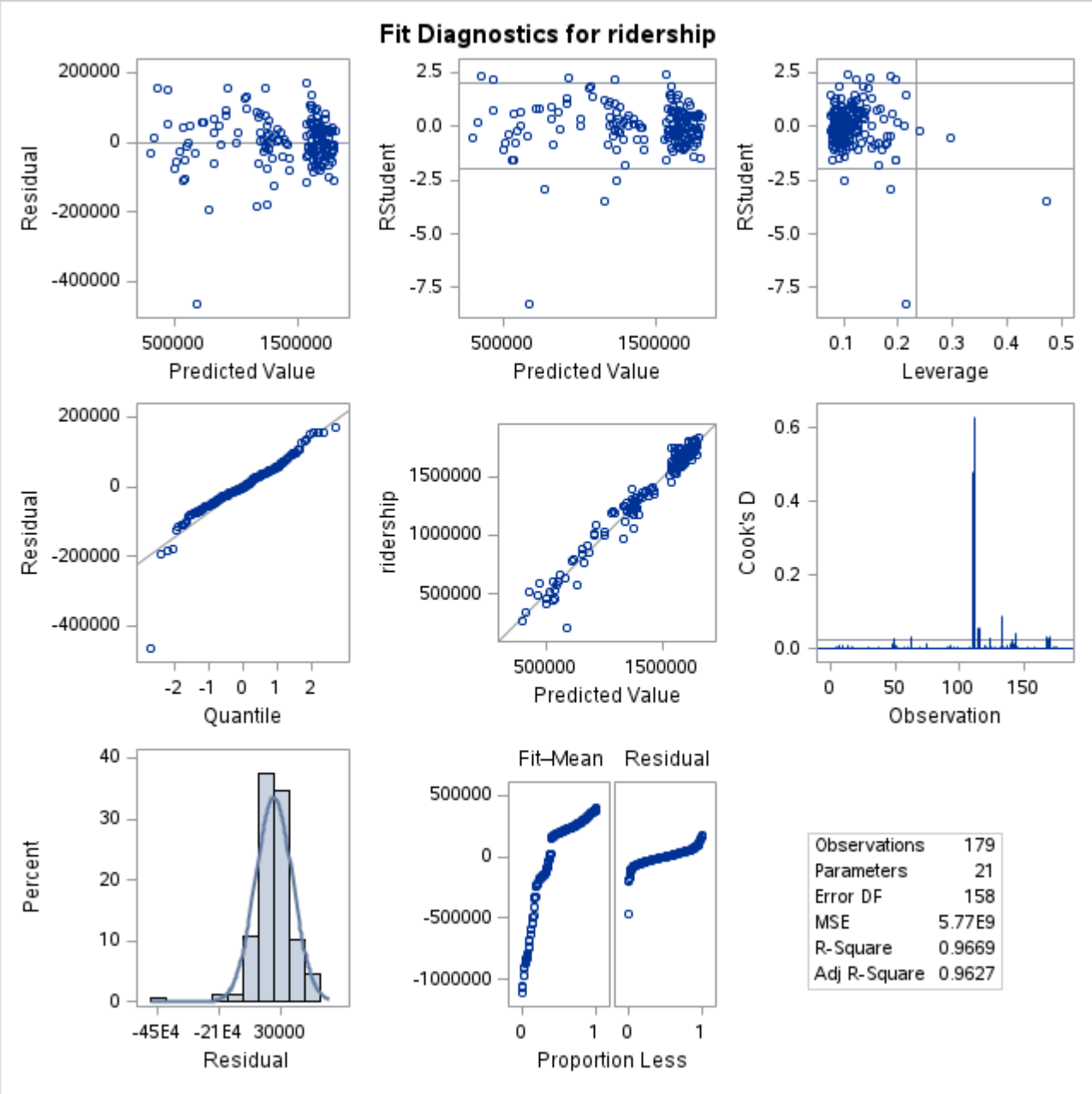
Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	20	2.66369E13	1.331845E12	230.67	<.0001
Error	158	9.122673E11	5773843664		
Corrected Total	178	2.754917E13			

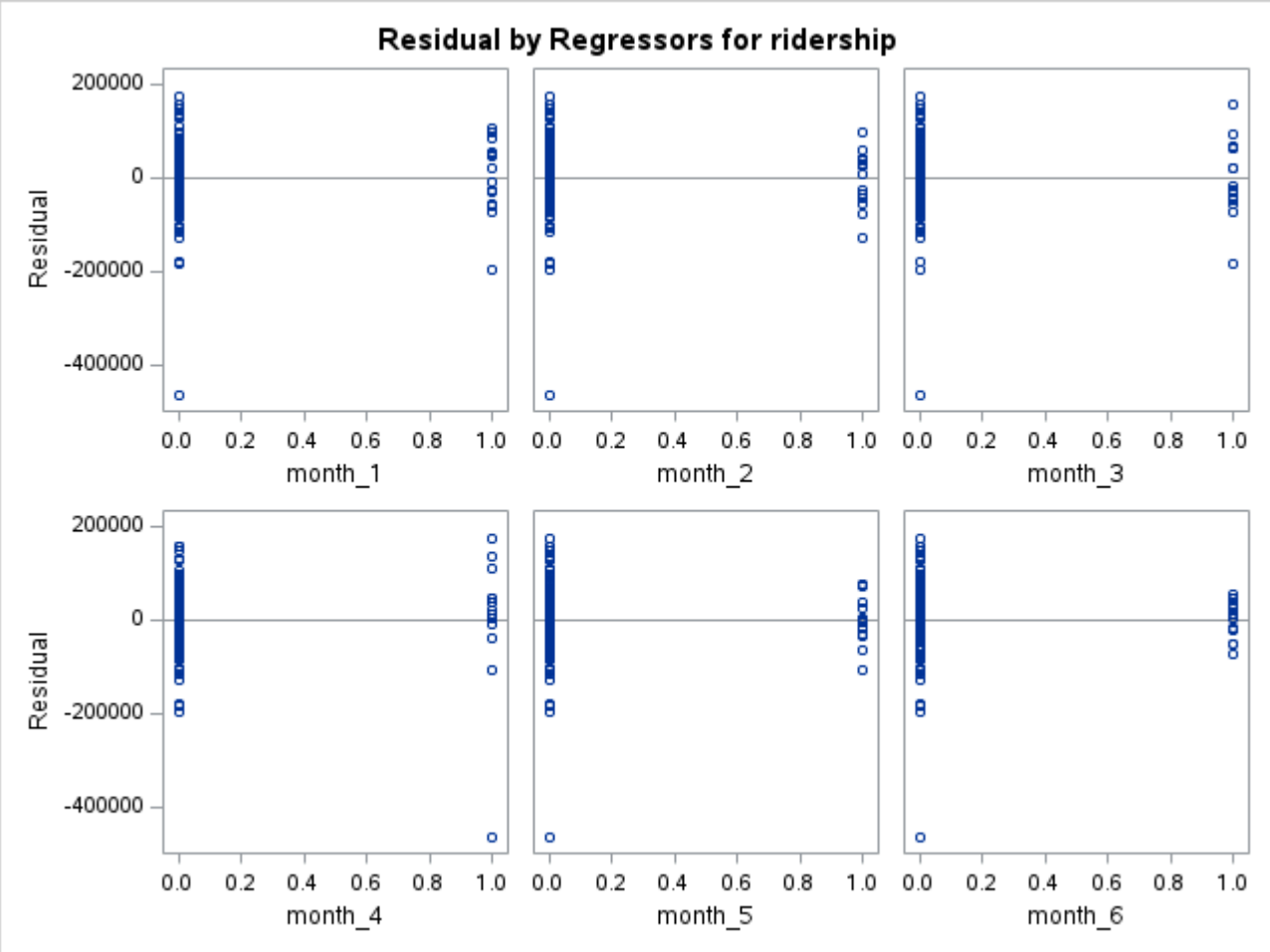
Root MSE	75986	R-Square	0.9669
Dependent Mean	1406726	Adj R-Sq	0.9627
Coeff Var	5.40161		

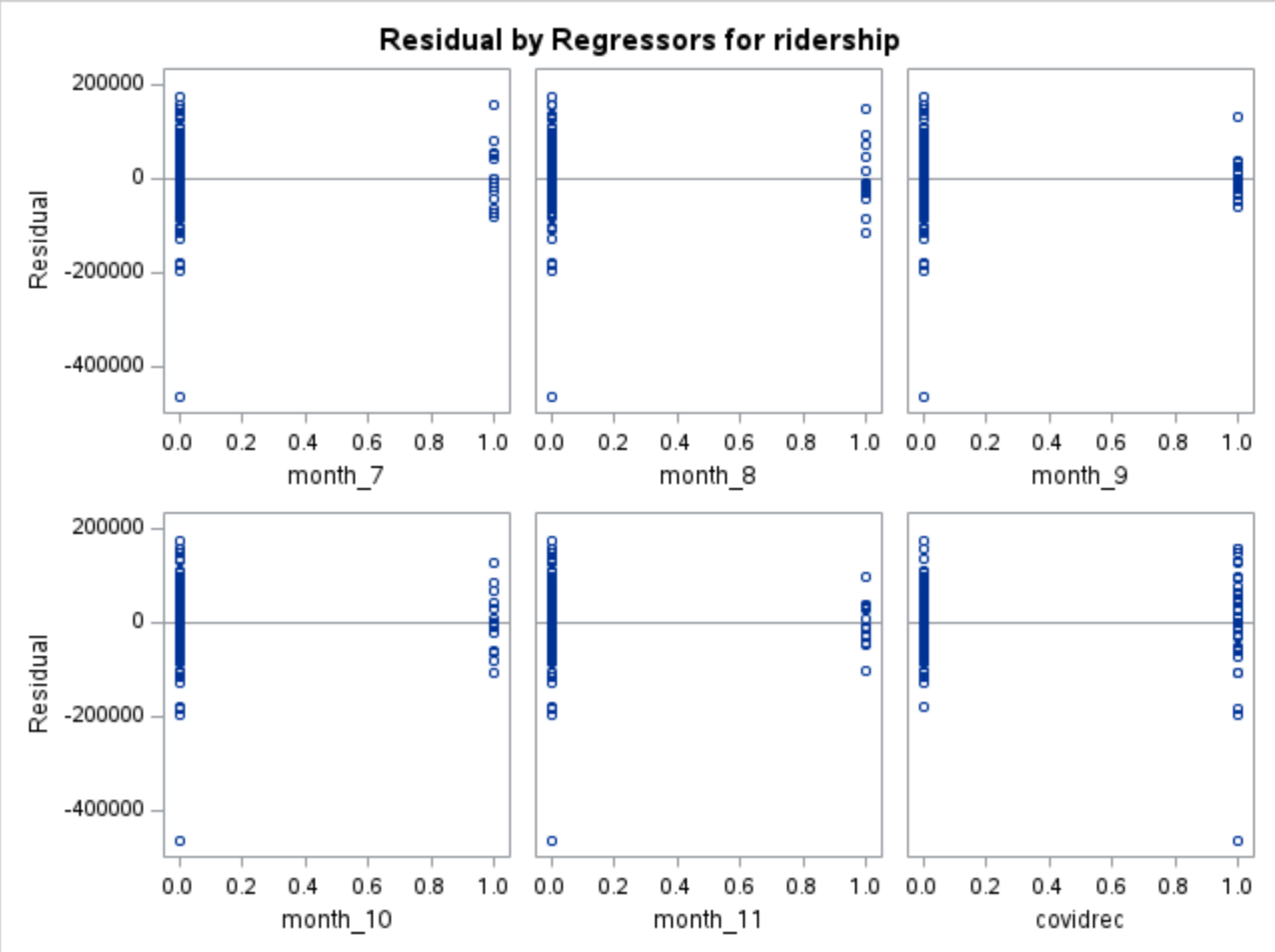
Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	551150	127571	4.32	<.0001	.	0
month_1	1	98521	31957	3.08	0.0024	0.43809	2.28265
month_2	1	134496	31510	4.27	<.0001	0.42313	2.36333
month_3	1	95644	28850	3.32	0.0011	0.50476	1.98114
month_4	1	42341	36693	1.15	0.2503	0.31204	3.20468
month_5	1	73610	54332	1.35	0.1774	0.14232	7.02637
month_6	1	60147	68736	0.88	0.3829	0.08892	11.24568
month_7	1	18784	78168	0.24	0.8104	0.06876	14.54384
month_8	1	14717	74666	0.20	0.8440	0.07536	13.26968
month_9	1	206306	64546	3.20	0.0017	0.10084	9.91658
month_10	1	138819	45115	3.08	0.0025	0.20642	4.84457
month_11	1	143577	31901	4.50	<.0001	0.41284	2.42224
covidrec	1	-532622	50055	-10.64	<.0001	0.08367	11.95162
postcovid	1	-281217	30221	-9.31	<.0001	0.21981	4.54932
ridership_lag1	1	0.50601	0.05580	9.07	<.0001	0.06735	14.84730
ridership_lag12	1	-0.03529	0.02988	-1.18	0.2393	0.23515	4.25262
gas_price_lag1	1	2258.84773	475.42178	4.75	<.0001	0.31309	3.19399
unemployment_rate_lag1	1	-3268.42656	5131.24738	-0.64	0.5251	0.45701	2.18812
avg_temp	1	960.11990	3408.32679	0.28	0.7785	0.02970	33.67318
total_precip_mm	1	-85.85014	178.39379	-0.48	0.6310	0.77669	1.28751
total_snow_cm	1	-156.10742	773.71787	-0.20	0.8404	0.22760	4.39369

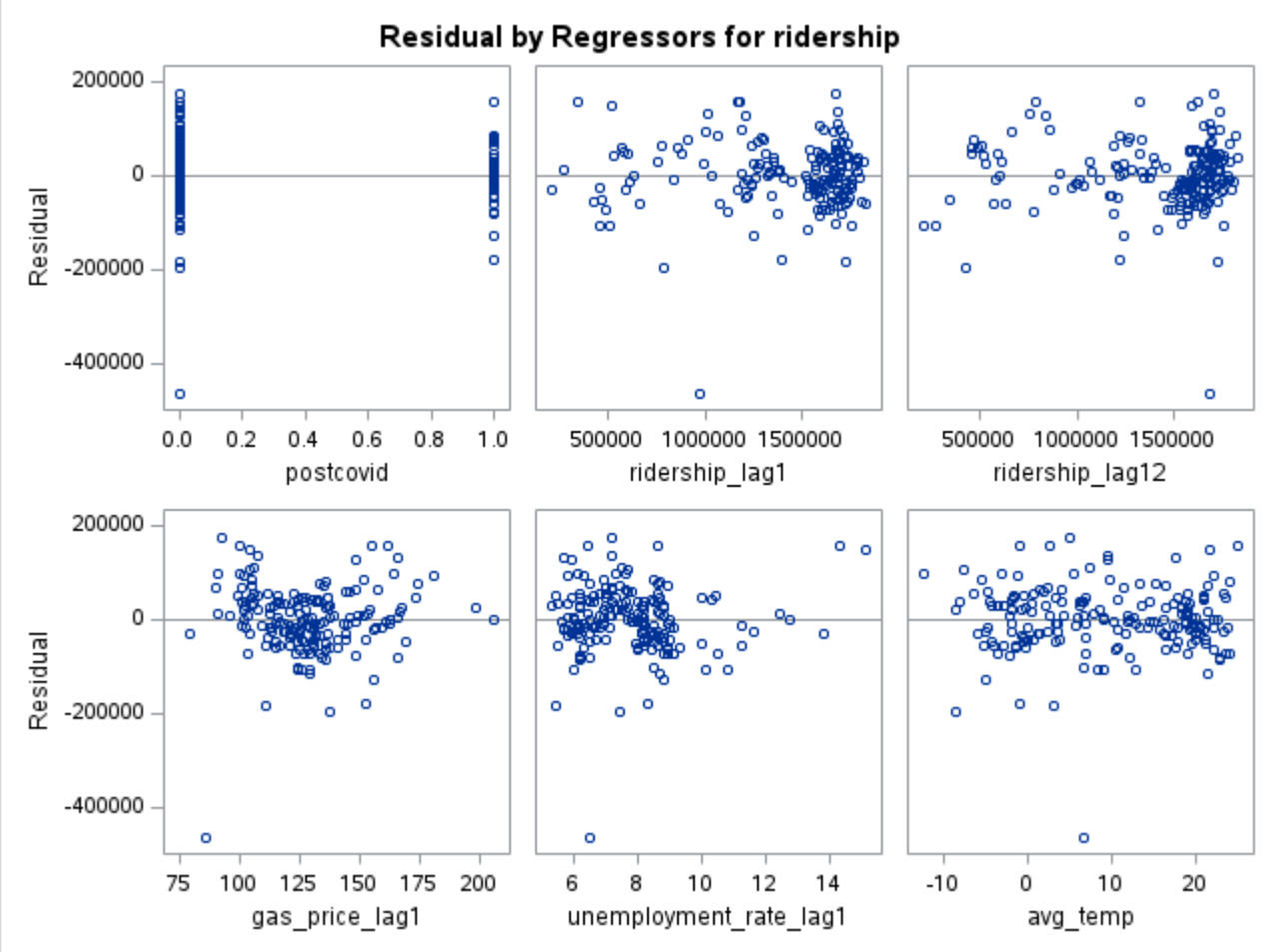
PROC REG Model 3: Full

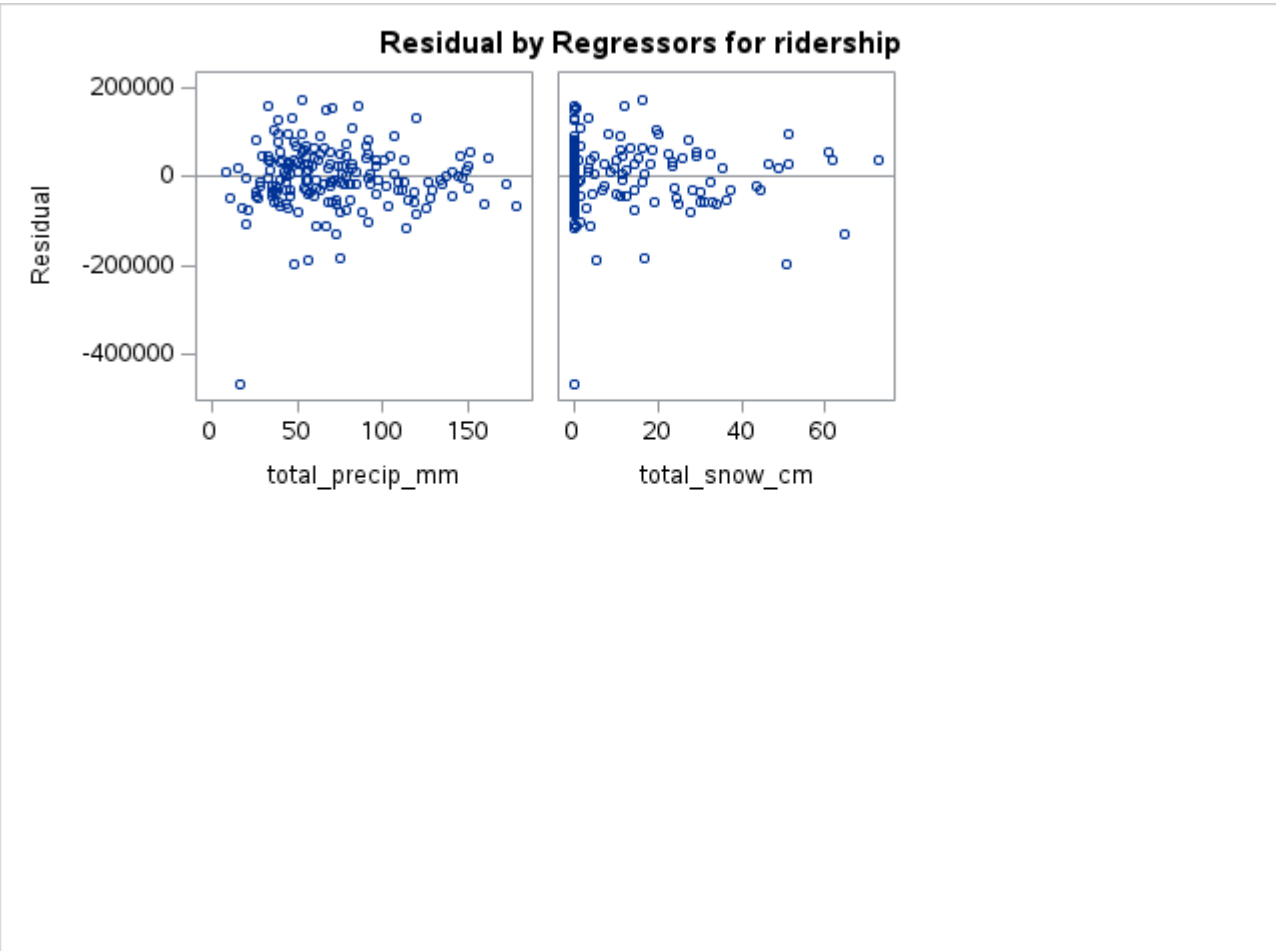
The REG Procedure
Model: MODEL1
Dependent Variable: ridership


Fit Diagnostics for ridership
Residual
Predicted Value
RStudent
Leverage
Quantile
ridership
Cook's D
Observation
Percent
Fit–Mean
Residual
Proportion Less
Observations 179
Parameters 21
Error DF 158
MSE 5.77E9
R-Square 0.9669
Adj R-Square 0.9627


Residual by Regressors for ridership
Residual
month_1
month_2
month_3
month_4
month_5
month_6


Residual by Regressors for ridership
Residual
200000
0
-200000
-400000
0.0 0.2 0.4 0.6 0.8 1.0
month_7
month_8
month_9
month_10
month_11
covidrec


Residual by Regressors for ridership
Residual
200000
0
-200000
-400000
0.0 0.2 0.4 0.6 0.8 1.0
postcovid
500000 1000000 1500000
ridership_lag1
ridership_lag12
75 100 125 150 175 200
gas_price_lag1
6 8 10 12 14
unemployment_rate_lag1
-10 0 10 20
avg_temp


Residual by Regressors for ridership
Residual
200000
0
-200000
-400000
0 50 100 150
total_precip_mm
0 20 40 60
total_snow_cm

PROC REG Model 1: Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	109
Number of Observations Used	109

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	11	4.124335E11	37493954545	17.59	<.0001
Error	97	2.067225E11	2131159794		
Corrected Total	108	6.19156E11			

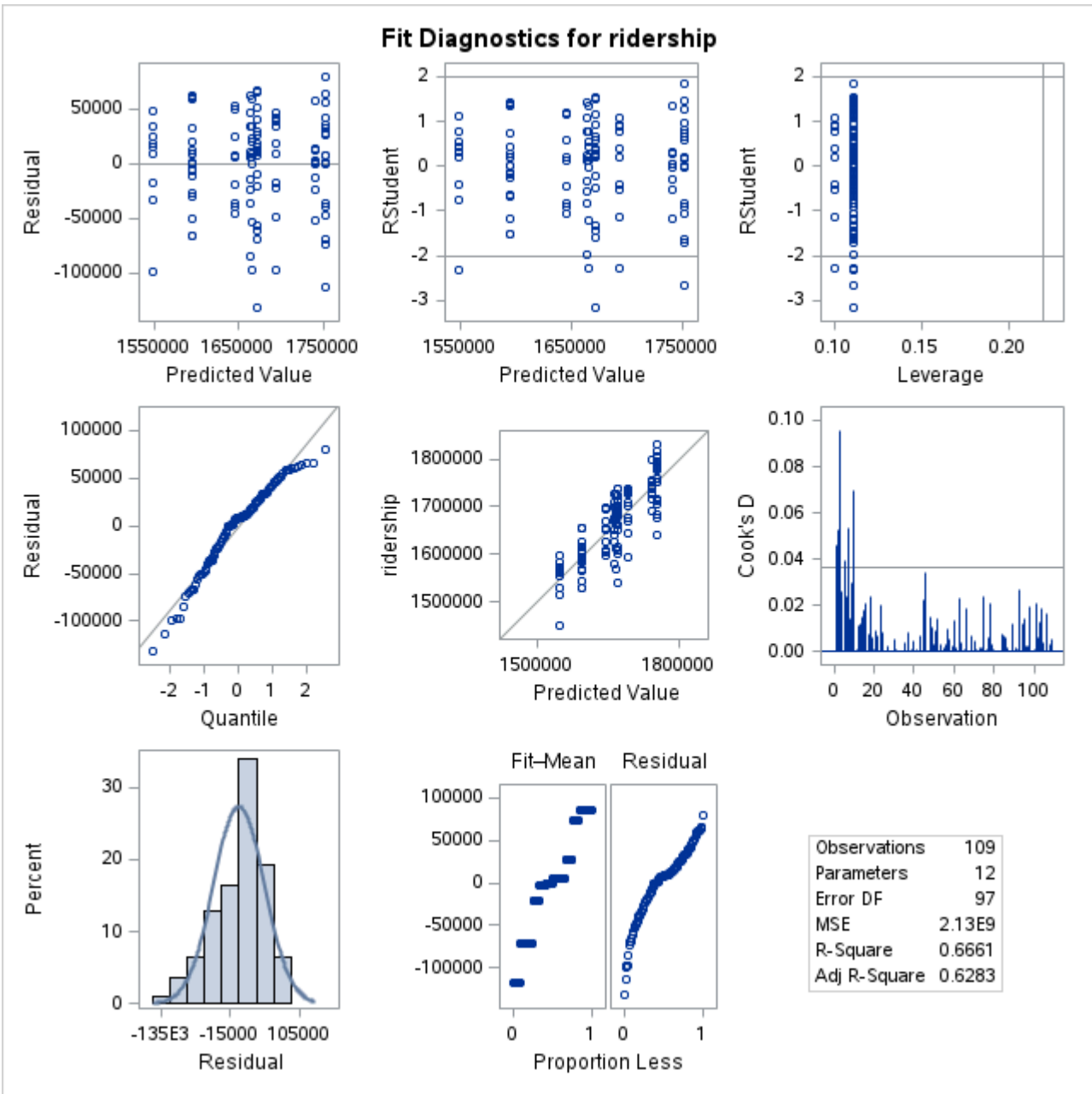
Root MSE	46164	**R-Square**	0.6661
Dependent Mean	1666000	**Adj R-Sq**	0.6283
Coeff Var	2.77098		

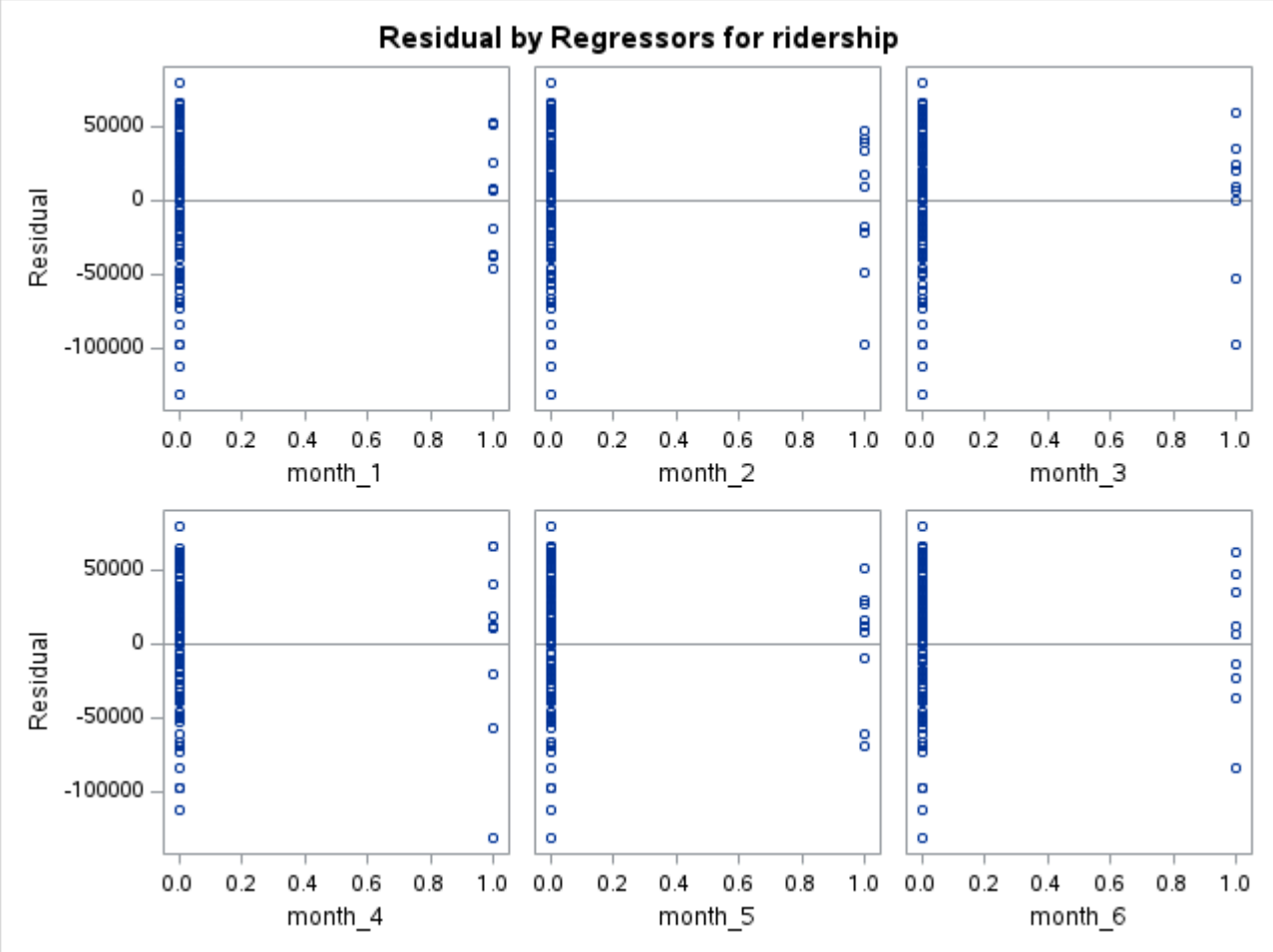
Parameter Estimates

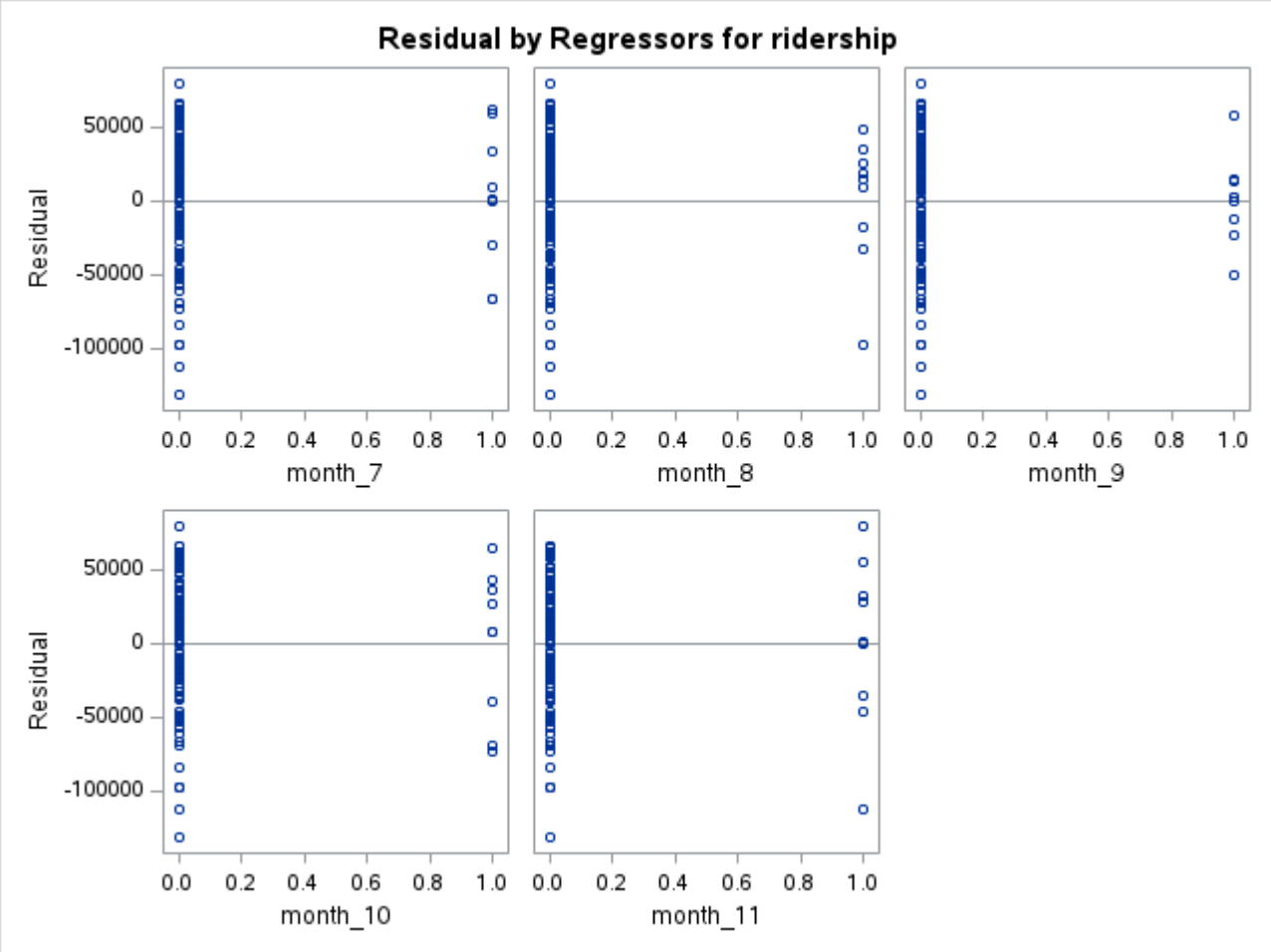
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	1594444	15388	103.62	<.0001	.	0
month_1	1	50444	21762	2.32	0.0226	0.54500	1.83486
month_2	1	98056	21211	4.62	<.0001	0.52153	1.91743
month_3	1	71333	21762	3.28	0.0015	0.54500	1.83486
month_4	1	76556	21762	3.52	0.0007	0.54500	1.83486
month_5	1	76556	21762	3.52	0.0007	0.54500	1.83486
month_6	1	69333	21762	3.19	0.0019	0.54500	1.83486
month_7	1	-222.22222	21762	-0.01	0.9919	0.54500	1.83486
month_8	1	-46889	21762	-2.15	0.0337	0.54500	1.83486
month_9	1	145778	21762	6.70	<.0001	0.54500	1.83486
month_10	1	157444	21762	7.23	<.0001	0.54500	1.83486
month_11	1	157333	21762	7.23	<.0001	0.54500	1.83486

PROC REG Model 1: Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership







PROC REG Model 2: Lag Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	109
Number of Observations Used	109

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	13	5.332671E11	41020545672	45.37	<.0001
Error	95	85888906267	904093750		
Corrected Total	108	6.19156E11			

Root MSE	30068	**R-Square**	0.8613
Dependent Mean	1666000	**Adj R-Sq**	0.8423
Coeff Var	1.80481		

Parameter Estimates

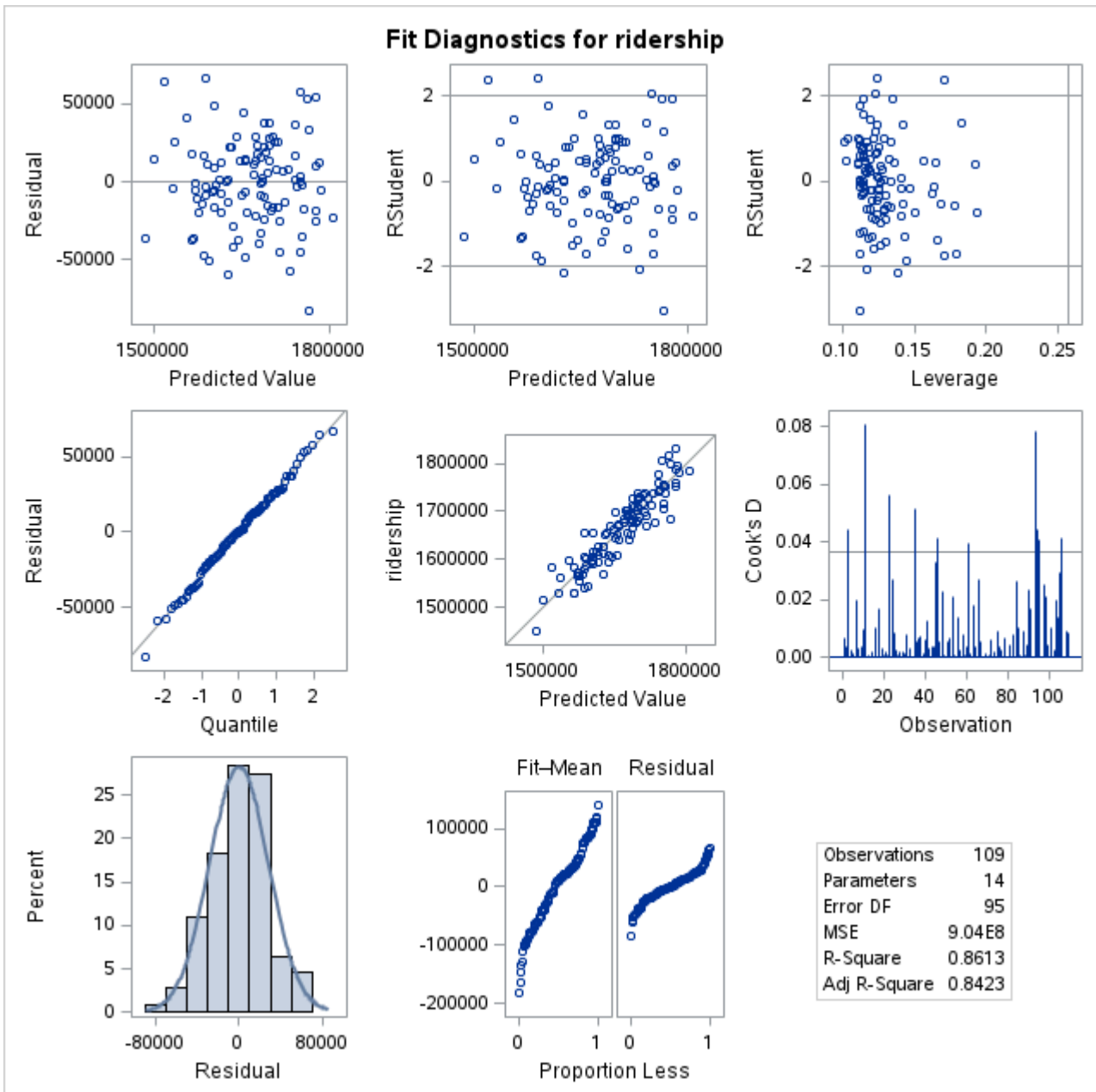
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	562710	114205	4.93	<.0001	.	0
month_1	1	60972	21932	2.78	0.0066	0.22763	4.39301
month_2	1	81969	20804	3.94	0.0002	0.23000	4.34786
month_3	1	58743	16727	3.51	0.0007	0.39135	2.55527

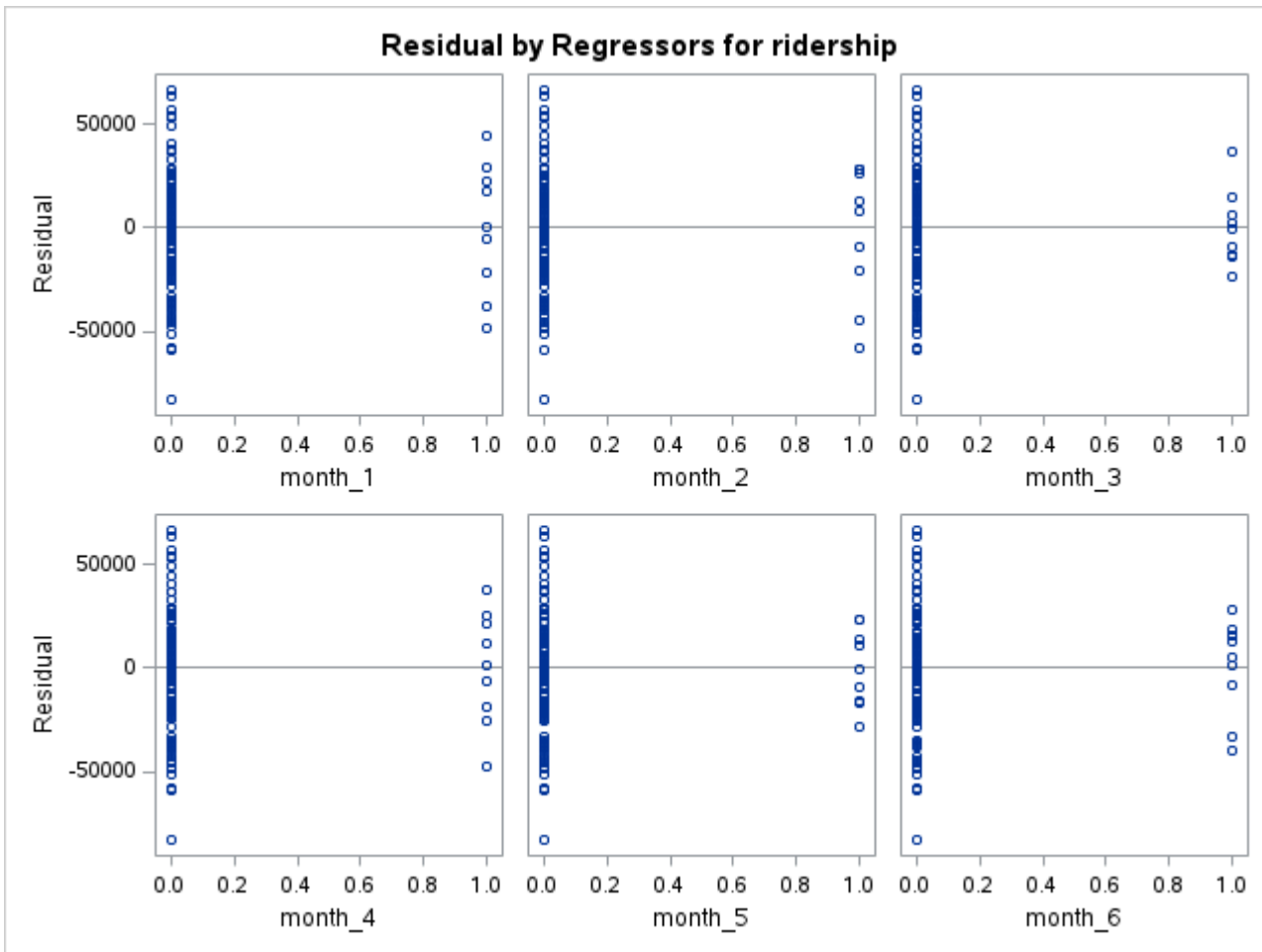
Parameter Estimates

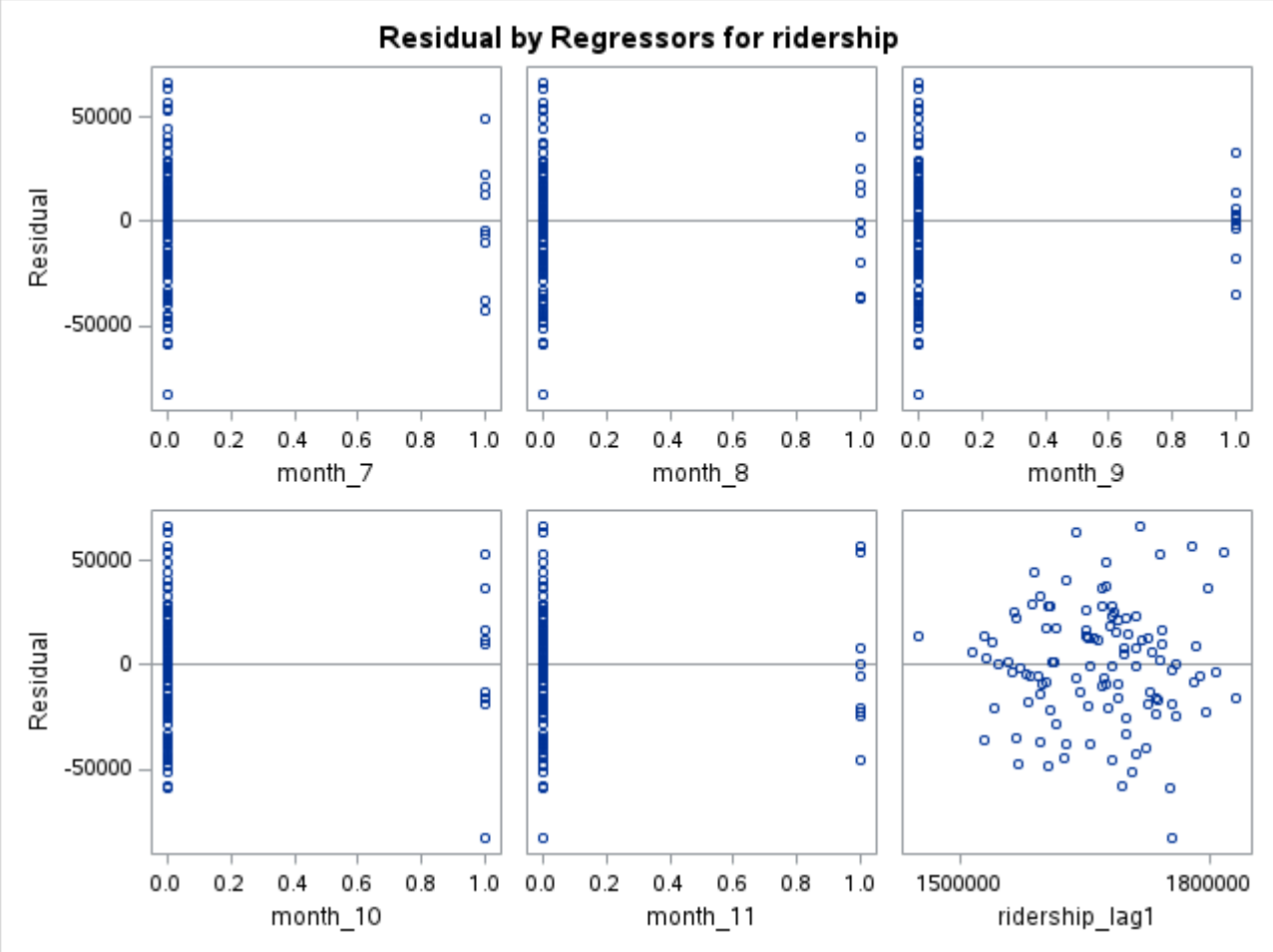
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
month_4	1	63814	18268	3.49	0.0007	0.32812	3.04770
month_5	1	61421	18097	3.39	0.0010	0.33435	2.99092
month_6	1	61079	17539	3.48	0.0008	0.35594	2.80945
month_7	1	18948	16201	1.17	0.2451	0.41716	2.39718
month_8	1	7808.97090	18649	0.42	0.6764	0.31483	3.17630
month_9	1	127669	29396	4.34	<.0001	0.12671	7.89200
month_10	1	93803	17858	5.25	<.0001	0.34336	2.91243
month_11	1	96580	16948	5.70	<.0001	0.38122	2.62313
ridership_lag1	1	0.20382	0.09066	2.25	0.0269	0.17431	5.73692
ridership_lag12	1	0.42701	0.06537	6.53	<.0001	0.26218	3.81424

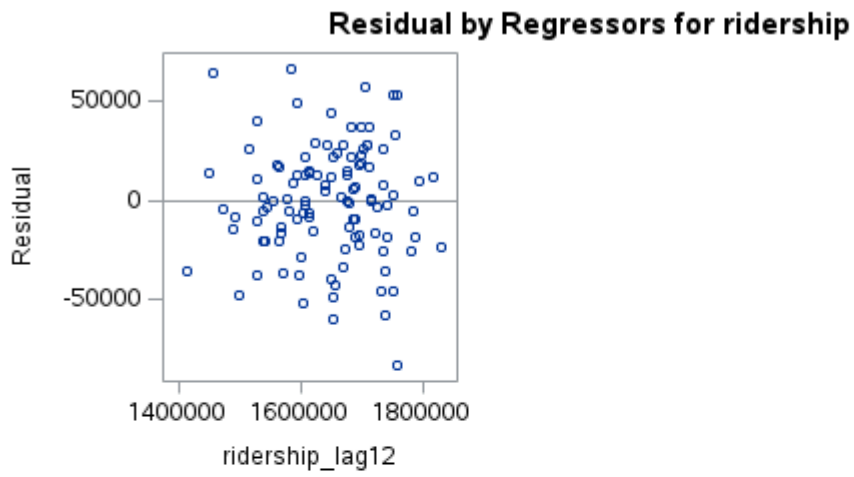
PROC REG Model 2: Lag Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership









Pre-Covid PROC REG Model 3

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	109
Number of Observations Used	109

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	18	5.35933E11	29774054362	32.20	<.0001
Error	90	83223021493	924700239		
Corrected Total	108	6.19156E11			

Root MSE	30409	**R-Square**	0.8656
Dependent Mean	1666000	**Adj R-Sq**	0.8387
Coeff Var	1.82526		

Parameter Estimates

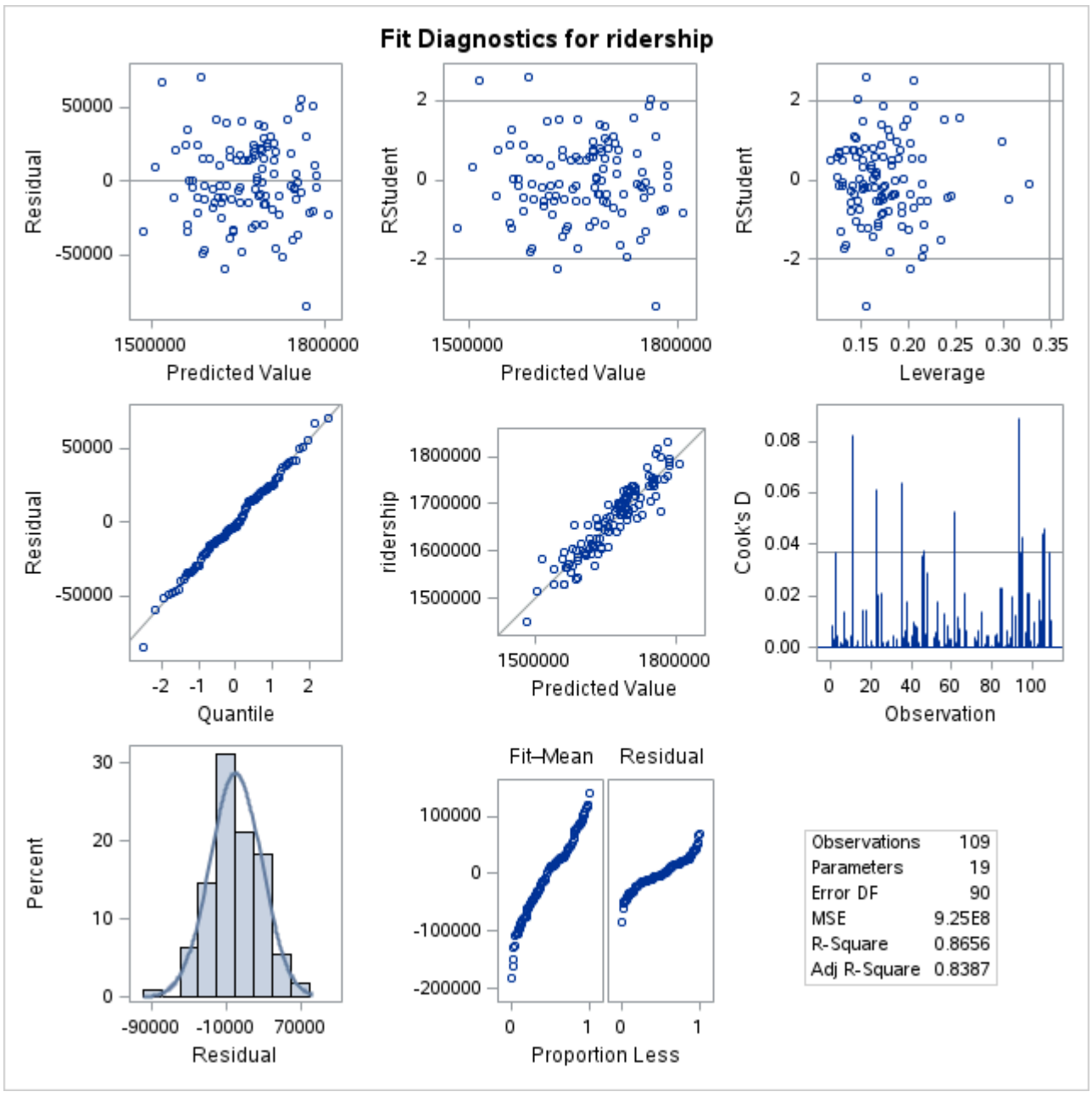
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	603743	168964	3.57	0.0006	.	0
month_1	1	59382	23088	2.57	0.0117	0.21010	4.75969
month_2	1	75462	21802	3.46	0.0008	0.21420	4.66857
month_3	1	62252	17744	3.51	0.0007	0.35569	2.81141

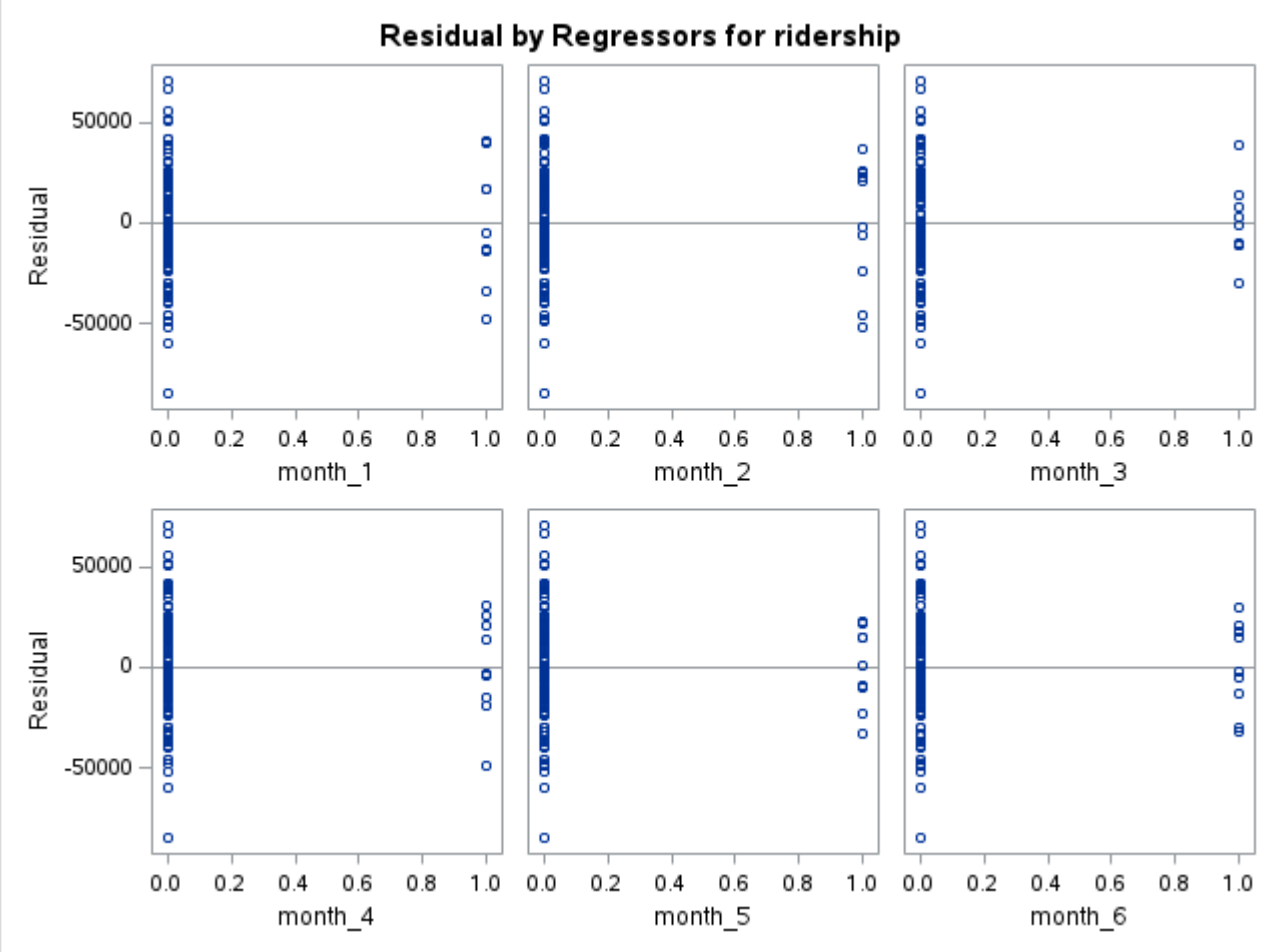
Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
month_4	1	78139	22162	3.53	0.0007	0.22801	4.38575
month_5	1	80308	30474	2.64	0.0099	0.12059	8.29240
month_6	1	83691	35915	2.33	0.0220	0.08682	11.51788
month_7	1	40134	39172	1.02	0.3083	0.07299	13.70123
month_8	1	27716	37942	0.73	0.4670	0.07779	12.85443
month_9	1	146922	41472	3.54	0.0006	0.06511	15.35799
month_10	1	113173	26936	4.20	<.0001	0.15436	6.47856
month_11	1	105636	19837	5.33	<.0001	0.28459	3.51385
ridership_lag1	1	0.19858	0.09315	2.13	0.0357	0.16887	5.92167
ridership_lag12	1	0.41443	0.07498	5.53	<.0001	0.20384	4.90577
gas_price_lag1	1	-139.87035	297.45381	-0.47	0.6393	0.68953	1.45026
unemployment_rate_lag1	1	354.62071	3561.00228	0.10	0.9209	0.58417	1.71184
avg_temp	1	-369.80594	1733.81881	-0.21	0.8316	0.02894	34.55304
total_precip_mm	1	-122.05665	90.60339	-1.35	0.1813	0.76035	1.31518
total_snow_cm	1	332.50407	422.39387	0.79	0.4332	0.17755	5.63233

Pre-Covid PROC REG Model 3

The REG Procedure
Model: MODEL1
Dependent Variable: ridership




Residual by Regressors for ridership
Residual
50000
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-50000
0.0 0.2 0.4 0.6 0.8 1.0
month_1
month_2
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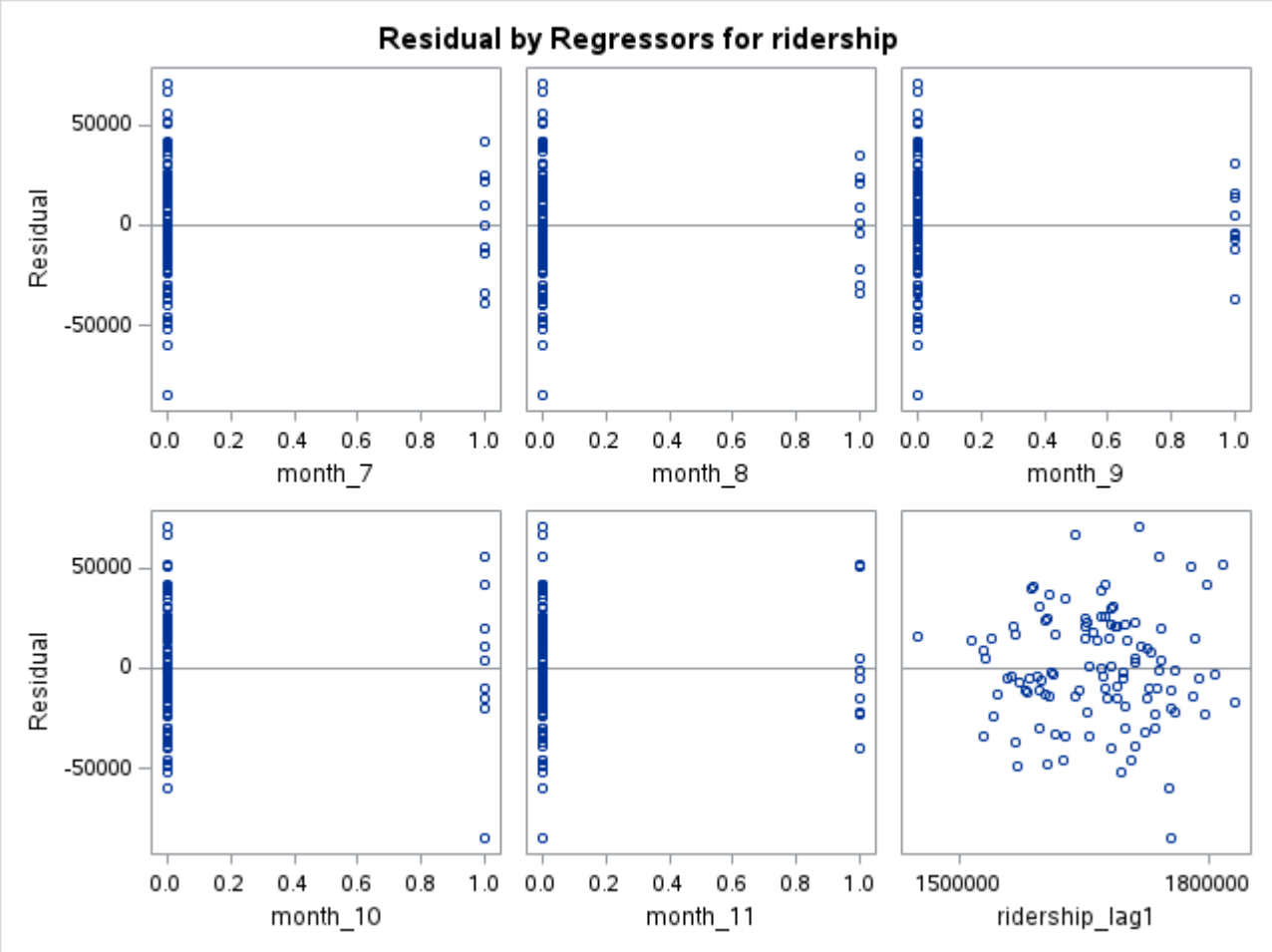
month_4
month_5
month_6


Residual by Regressors for ridership
Residual
50000
0
-50000
0.0 0.2 0.4 0.6 0.8 1.0
month_7
month_8
month_9
month_10
month_11
1500000 1800000
ridership_lag1

Residual by Regressors for ridership
Residual
50000
0
-50000
1400000 1600000 1800000
ridership_lag12
100 120 140
gas_price_lag1
5 6 7 8 9
unemployment_rate_lag1
-10 0 10 20
avg_temp
0 50 100 150
total_precip_mm
0 20 40 60
total_snow_cm

Pre-Covid PROC REG Model 3

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	109
Number of Observations Used	109

Stepwise Selection: Step 1

Variable ridership_lag12 Entered: R-Square = 0.7716 and C(p) = 47.9240

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	4.777471E11	4.777471E11	361.50	<.0001
Error	107	1.414089E11	1321578120		
Corrected Total	108	6.19156E11			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	399571	66699	47428536794	35.89	<.0001
ridership_lag12	0.76946	0.04047	4.777471E11	361.50	<.0001

Bounds on condition number: 1, 1

Stepwise Selection: Step 2

Variable unemployment_rate_lag1 Entered: R-Square = 0.7863 and C(p) = 40.1048

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	4.868269E11	2.434135E11	194.98	<.0001
Error	106	1.323291E11	1248387517		
Corrected Total	108	6.19156E11			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	241977	87276	9596328672	7.69	0.0066
ridership_lag12	0.82236	0.04395	4.370188E11	350.07	<.0001
unemployment_rate_lag1	9530.19931	3533.77569	9079782113	7.27	0.0081

Bounds on condition number: 1.2487, 4.9947

Stepwise Selection: Step 3

Variable month_11 Entered: R-Square = 0.7964 and C(p) = 35.3192

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	4.931016E11	1.643672E11	136.91	<.0001
Error	105	1.260544E11	1200518396		
Corrected Total	108	6.19156E11			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	292249	88366	13131142021	10.94	0.0013
month_11	28726	12565	6274645246	5.23	0.0243
ridership_lag12	0.79374	0.04488	3.754597E11	312.75	<.0001
unemployment_rate_lag1	8781.42764	3480.80562	7640812296	6.36	0.0131

Bounds on condition number: 1.354, 11.099

Stepwise Selection: Step 4

Variable month_8 Entered: R-Square = 0.8057 and C(p) = 31.0693

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	4	4.988809E11	1.247202E11	107.84	<.0001
Error	104	1.202751E11	1156491489		
Corrected Total	108	6.19156E11			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	393868	97922	18710578358	16.18	0.0001
month_8	-29993	13417	5779316648	5.00	0.0275
month_11	30037	12347	6844970112	5.92	0.0167
ridership_lag12	0.74214	0.04973	2.575403E11	222.69	<.0001
unemployment_rate_lag1	6844.86014	3524.50575	4361894439	3.77	0.0548

Bounds on condition number: 1.7257, 21.76

Stepwise Selection: Step 5

Variable month_10 Entered: R-Square = 0.8138 and C(p) = 27.6608

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	5.038822E11	1.007764E11	90.05	<.0001
Error	103	1.152738E11	1119163412		
Corrected Total	108	6.19156E11			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	468657	102620	23342236035	20.86	<.0001
month_8	-32508	13252	6734286034	6.02	0.0158
month_10	26758	12658	5001283483	4.47	0.0369
month_11	35477	12415	9138594753	8.17	0.0052
ridership_lag12	0.70081	0.05269	1.980165E11	176.93	<.0001
unemployment_rate_lag1	5600.88400	3516.74257	2838739256	2.54	0.1143

Bounds on condition number: 2.0013, 34.979

Stepwise Selection: Step 6

Variable month_9 Entered: R-Square = 0.8260 and C(p) = 21.5294

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	6	5.114012E11	85233536259	80.68	<.0001
Error	102	1.077548E11	1056419436		
Corrected Total	108	6.19156E11			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	568732	106525	30112813916	28.50	<.0001
month_8	-35325	12919	7898938386	7.48	0.0074
month_9	33093	12404	7519048950	7.12	0.0089
month_10	35384	12716	8179761640	7.74	0.0064
month_11	43361	12419	12878312587	12.19	0.0007
ridership_lag12	0.64520	0.05527	1.439677E11	136.28	<.0001
unemployment_rate_lag1	3923.34489	3474.11867	1347286813	1.28	0.2614

Bounds on condition number: 2.3332, 52.414

Stepwise Selection: Step 7

Variable unemployment_rate_lag1 Removed: R-Square = 0.8238 and C(p) = 20.9864

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	5.100539E11	1.020108E11	96.31	<.0001
Error	103	1.091021E11	1059243391		
Corrected Total	108	6.19156E11			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	653344	75824	78643716487	74.25	<.0001
month_8	-39207	12470	10471795329	9.89	0.0022
month_9	35628	12216	9010501392	8.51	0.0043
month_10	38330	12462	10020576358	9.46	0.0027
month_11	45757	12253	14772714083	13.95	0.0003
ridership_lag12	0.61123	0.04643	1.835771E11	173.31	<.0001

Bounds on condition number: 1.6422, 31.991

Stepwise Selection: Step 8

Variable month_7 Entered: R-Square = 0.8332 and C(p) = 16.6783

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	6	5.15887E11	85981172943	84.92	<.0001
Error	102	1.03269E11	1012440807		
Corrected Total	108	6.19156E11			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	702004	76852	84476731016	83.44	<.0001
month_7	-28055	11688	5833106919	5.76	0.0182
month_8	-45546	12474	13498270620	13.33	0.0004
month_9	34631	11950	8503160305	8.40	0.0046
month_10	37740	12186	9710409173	9.59	0.0025
month_11	44825	11985	14162183674	13.99	0.0003
ridership_lag12	0.58352	0.04684	1.571419E11	155.21	<.0001

Bounds on condition number: 1.7484, 46.072

All variables left in the model are significant at the 0.1500 level.

No other variable met the 0.0500 significance level for entry into the model.

Summary of Stepwise Selection								
Step	Variable Entered	Variable Removed	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	ridership_lag12		1	0.7716	0.7716	47.9240	361.50	<.0001
2	unemployment_rate_lag1		2	0.0147	0.7863	40.1048	7.27	0.0081
3	month_11		3	0.0101	0.7964	35.3192	5.23	0.0243
4	month_8		4	0.0093	0.8057	31.0693	5.00	0.0275
5	month_10		5	0.0081	0.8138	27.6608	4.47	0.0369
6	month_9		6	0.0121	0.8260	21.5294	7.12	0.0089
7		unemployment_rate_lag1	5	0.0022	0.8238	20.9864	1.28	0.2614
8	month_7		6	0.0094	0.8332	16.6783	5.76	0.0182

Pre-Covid PROC REG Model 3

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	109
Number of Observations Used	109

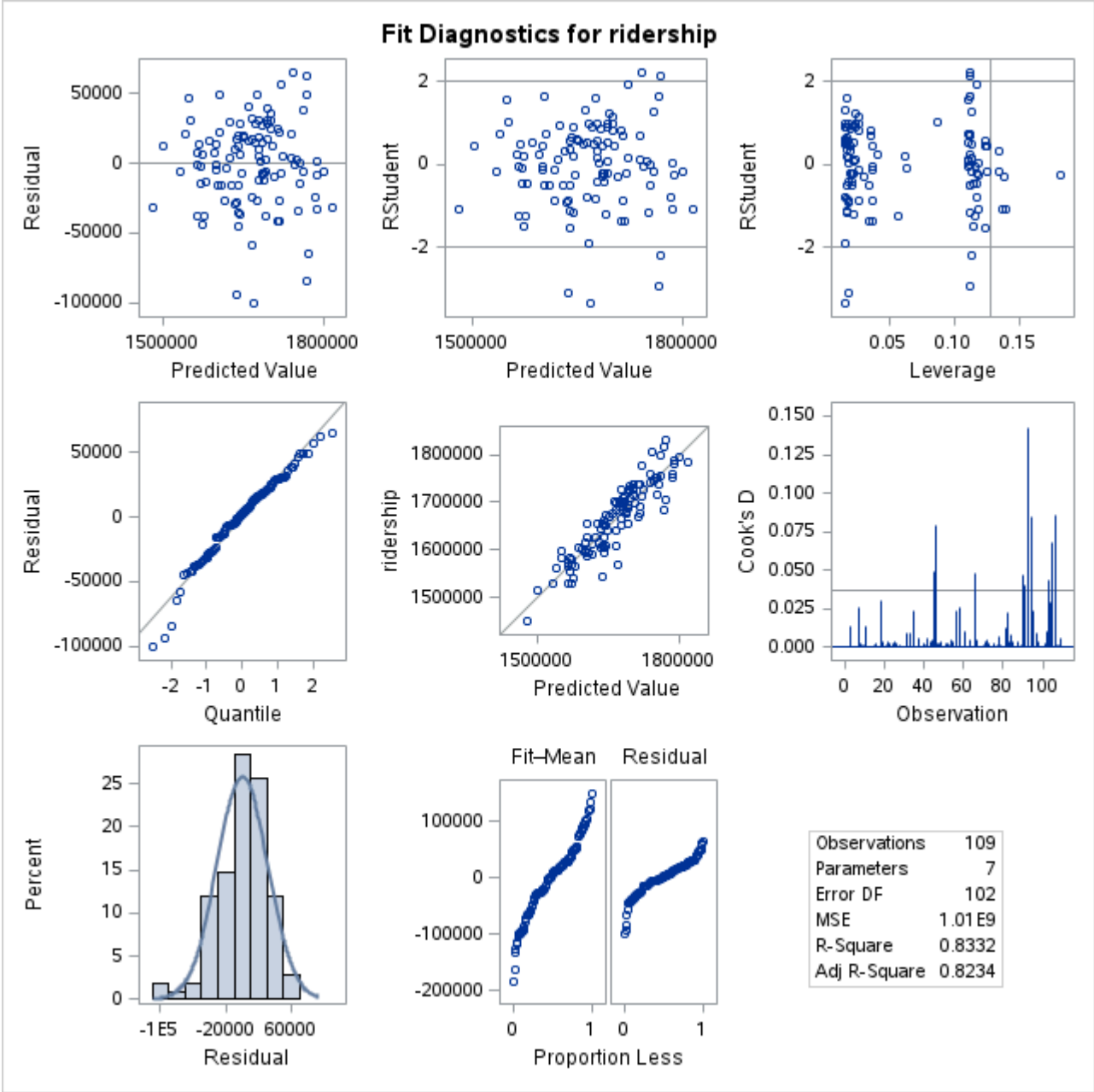
Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	6	5.15887E11	85981172943	84.92	<.0001
Error	102	1.03269E11	1012440807		
Corrected Total	108	6.19156E11			

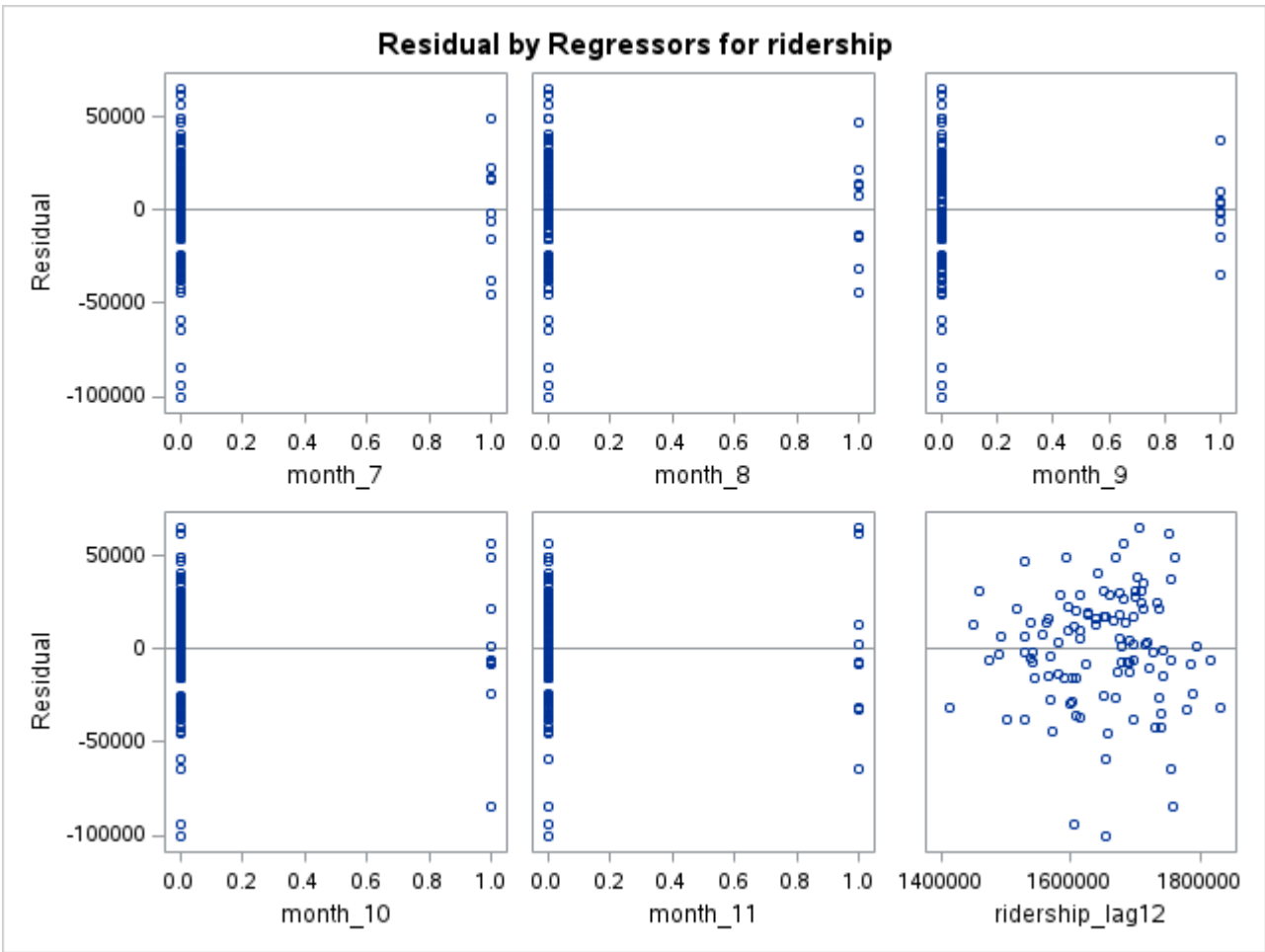
Root MSE	31819	R-Square	0.8332
Dependent Mean	1666000	Adj R-Sq	0.8234
Coeff Var	1.90990		

Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	702004	76852	9.13	<.0001	.	0
month_7	1	-28055	11688	-2.40	0.0182	0.89755	1.11414
month_8	1	-45546	12474	-3.65	0.0004	0.78807	1.26893
month_9	1	34631	11950	2.90	0.0046	0.85867	1.16459
month_10	1	37740	12186	3.10	0.0025	0.82570	1.21109
month_11	1	44825	11985	3.74	0.0003	0.85362	1.17148
ridership_lag12	1	0.58352	0.04684	12.46	<.0001	0.57194	1.74842

Pre-Covid PROC REG Model 3

The REG Procedure
Model: MODEL1
Dependent Variable: ridership





Pre-COVID Model P1: Month-only Seasonal Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	109
Number of Observations Used	109

Analysis of Variance					
Source	**DF**	**Sum of Squares**	**Mean Square**	**F Value**	**Pr > F**
Model	11	4.124335E11	37493954545	17.59	<.0001
Error	97	2.067225E11	2131159794		
Corrected Total	108	6.19156E11			

Root MSE	46164	**R-Square**	0.6661

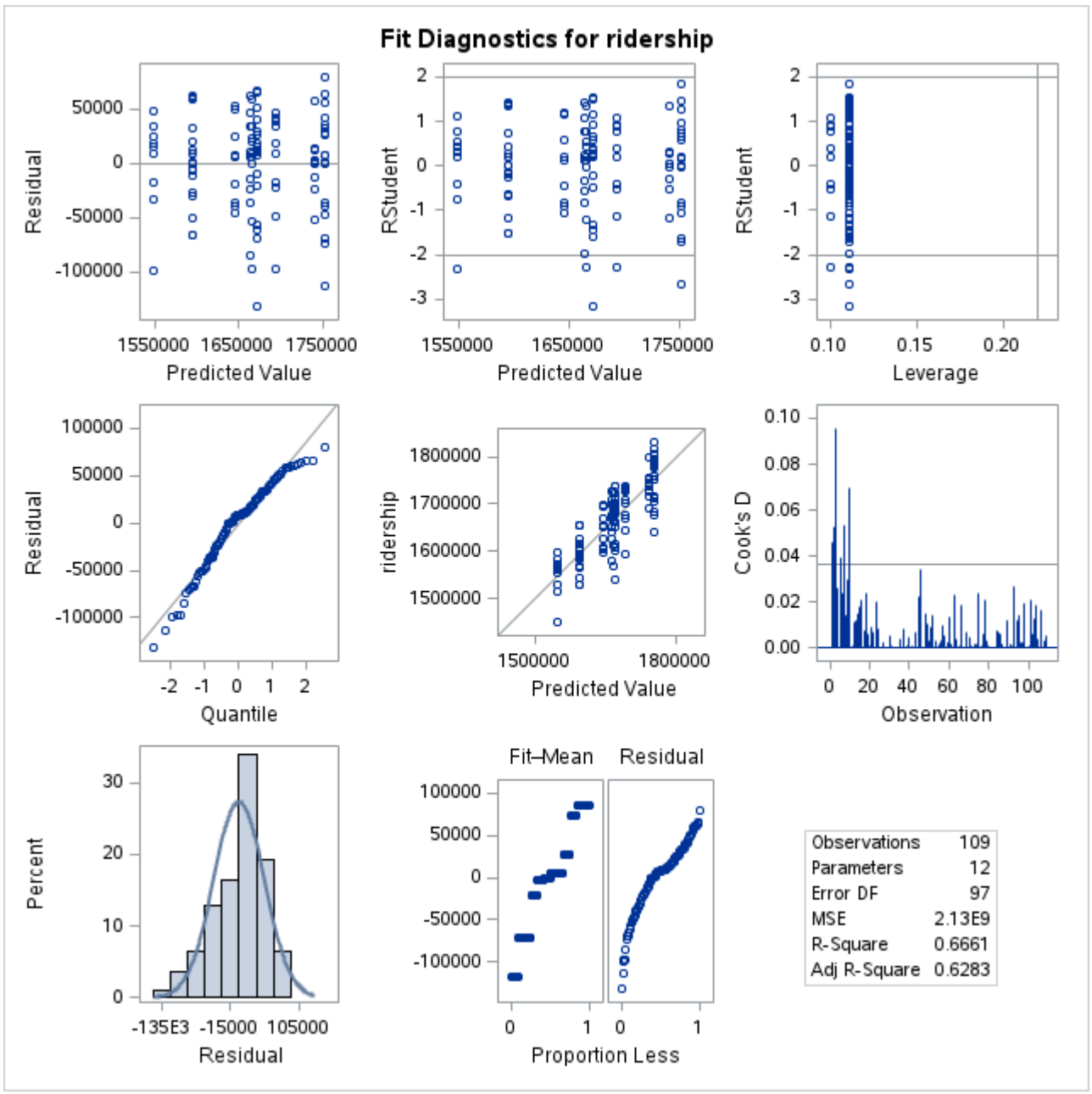
Dependent Mean	1666000	**Adj R-Sq**	0.6283
Coeff Var	2.77098		

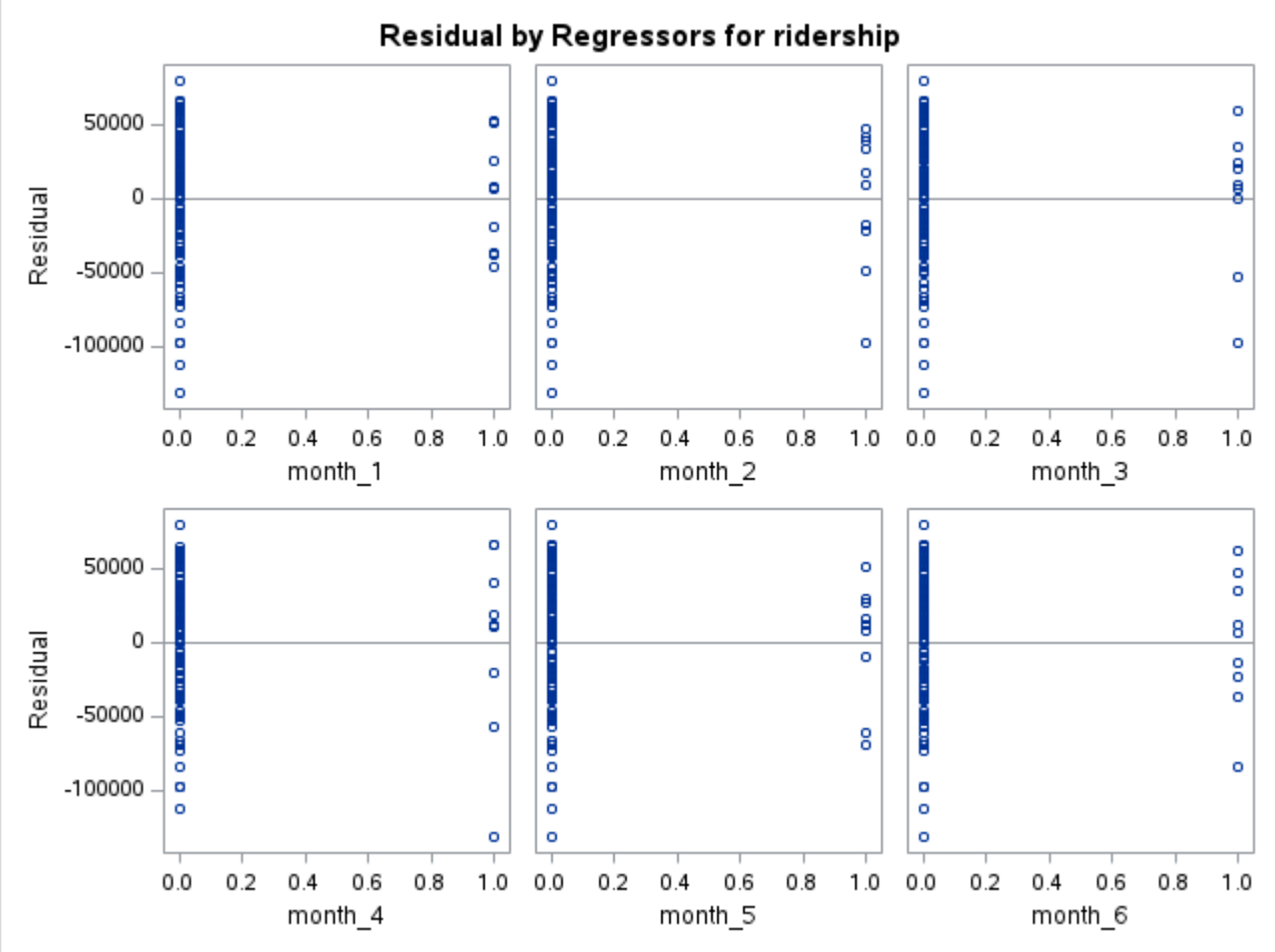
Parameter Estimates

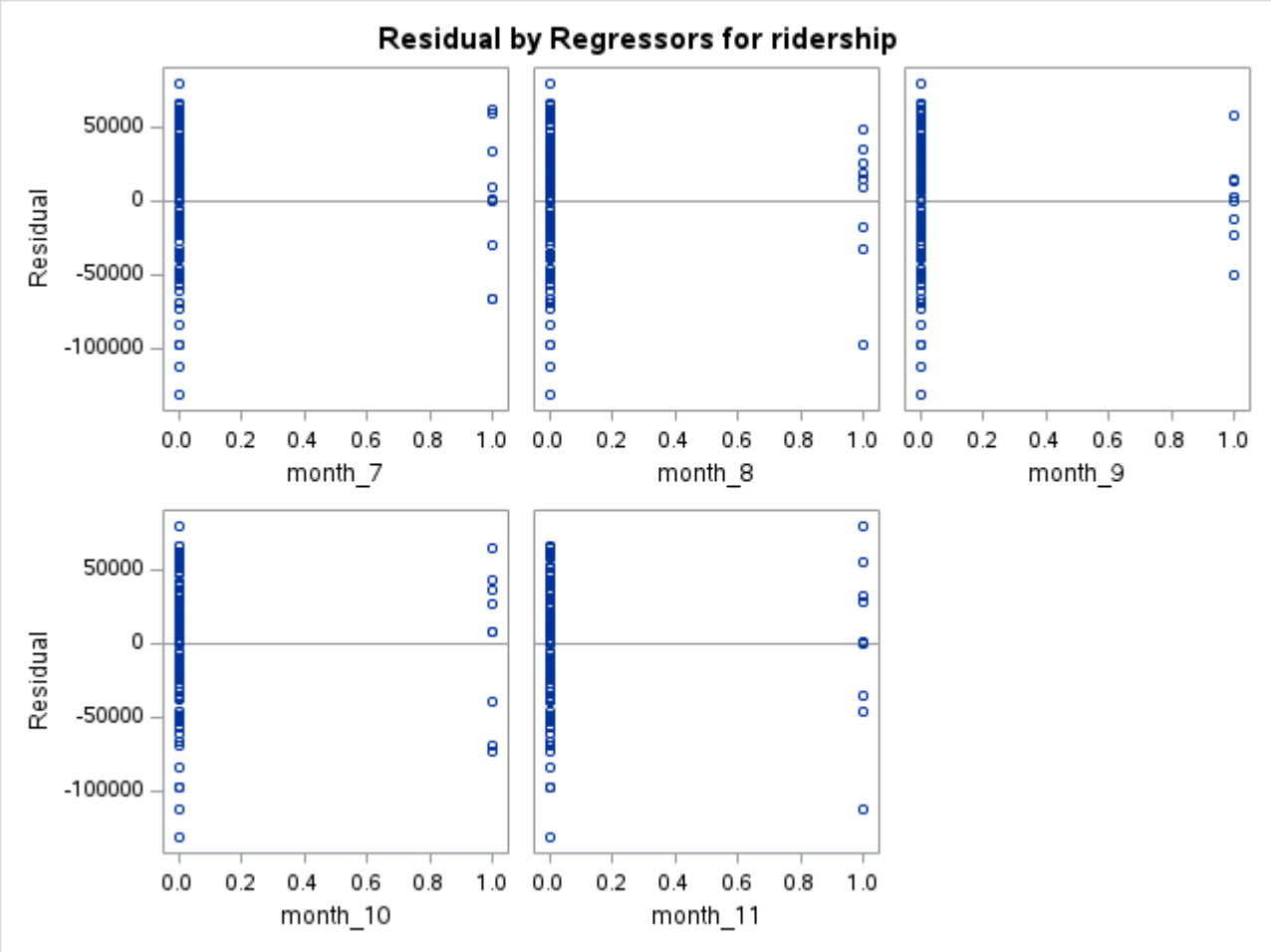
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	1594444	15388	103.62	<.0001	.	0
month_1	1	50444	21762	2.32	0.0226	0.54500	1.83486
month_2	1	98056	21211	4.62	<.0001	0.52153	1.91743
month_3	1	71333	21762	3.28	0.0015	0.54500	1.83486
month_4	1	76556	21762	3.52	0.0007	0.54500	1.83486
month_5	1	76556	21762	3.52	0.0007	0.54500	1.83486
month_6	1	69333	21762	3.19	0.0019	0.54500	1.83486
month_7	1	-222.22222	21762	-0.01	0.9919	0.54500	1.83486
month_8	1	-46889	21762	-2.15	0.0337	0.54500	1.83486
month_9	1	145778	21762	6.70	<.0001	0.54500	1.83486
month_10	1	157444	21762	7.23	<.0001	0.54500	1.83486
month_11	1	157333	21762	7.23	<.0001	0.54500	1.83486

Pre-COVID Model P1: Month-only Seasonal Baseline

The REG Procedure
Model: MODEL1
Dependent Variable: ridership







Pre-COVID Model P2: Lag Memory Model

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	109
Number of Observations Used	109

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	4.778188E11	2.389094E11	179.18	<.0001
Error	106	1.413372E11	1333369856		
Corrected Total	108	6.19156E11			

Root MSE	36515	**R-Square**	0.7717
Dependent Mean	1666000	**Adj R-Sq**	0.7674
Coeff Var	2.19180		

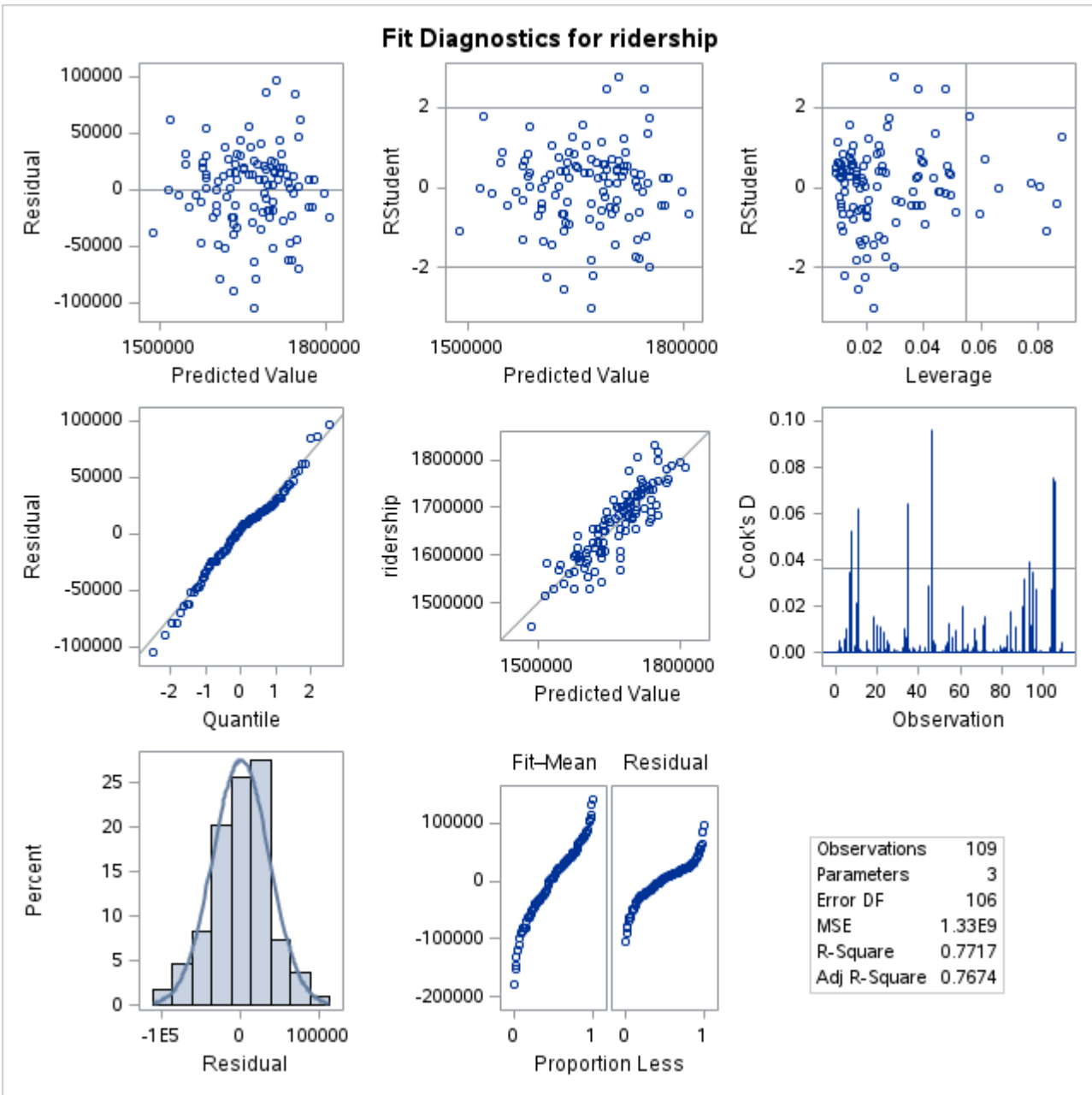
Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	412307	86641	4.76	<.0001	.	0
ridership_lag1	1	-0.01155	0.04982	-0.23	0.8171	0.85112	1.17492

Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
ridership_lag12	1	0.77340	0.04406	17.55	<.0001	0.85112	1.17492

Pre-COVID Model P2: Lag Memory Model

The REG Procedure
Model: MODEL1
Dependent Variable: ridership



Pre-COVID Model P3: Reduced External Model

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	109
Number of Observations Used	109

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	4	4.871199E11	1.2178E11	95.92	<.0001

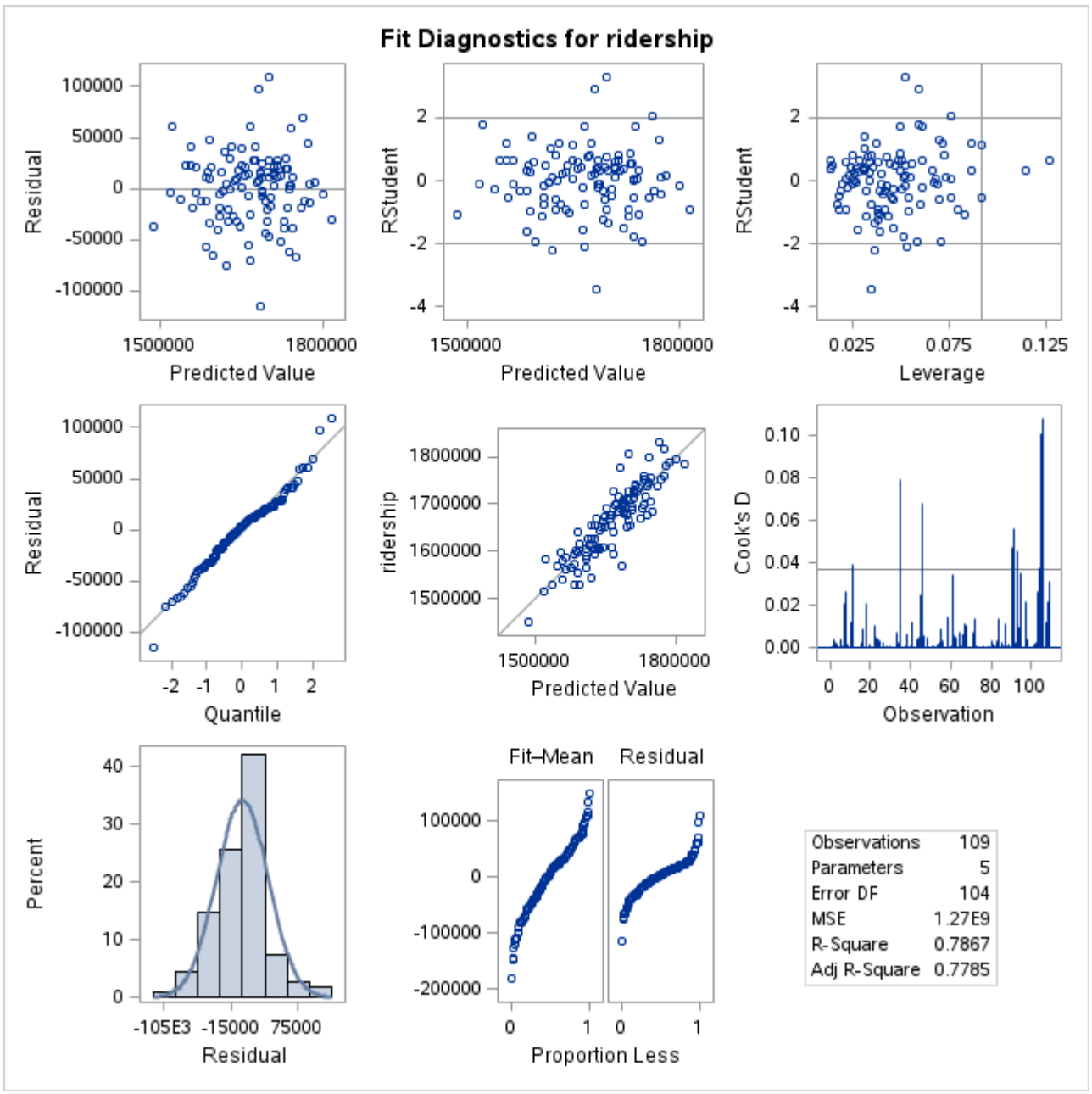
Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Error	104	1.320361E11	1269577551		
Corrected Total	108	6.19156E11			

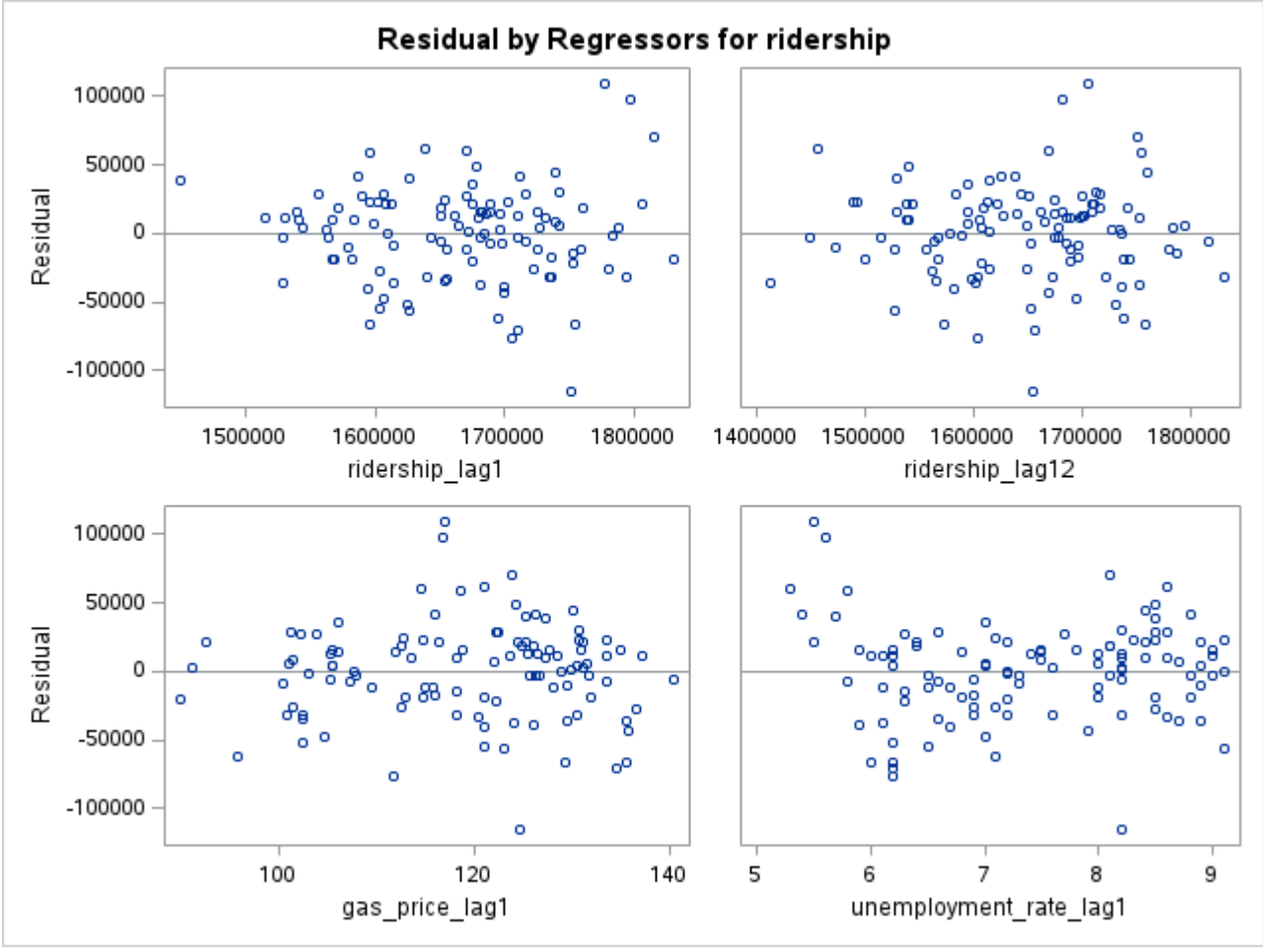
Root MSE	35631	R-Square	0.7867
Dependent Mean	1666000	Adj R-Sq	0.7785
Coeff Var	2.13872		

Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	208807	119721	1.74	0.0841	.	0
ridership_lag1	1	0.00988	0.04937	0.20	0.8418	0.82520	1.21183
ridership_lag12	1	0.82343	0.04698	17.53	<.0001	0.71282	1.40288
gas_price_lag1	1	141.85794	312.62610	0.45	0.6509	0.85704	1.16680
unemployment_rate_lag1	1	9271.45594	3681.06036	2.52	0.0133	0.75058	1.33231

Pre-COVID Model P3: Reduced External Model

The REG Procedure
Model: MODEL1
Dependent Variable: ridership





Pre-Covid Model P4: Whithout Ridership History

The REG Procedure
Model: MODEL1
Dependent Variable: ridership

Number of Observations Read	109
Number of Observations Used	109

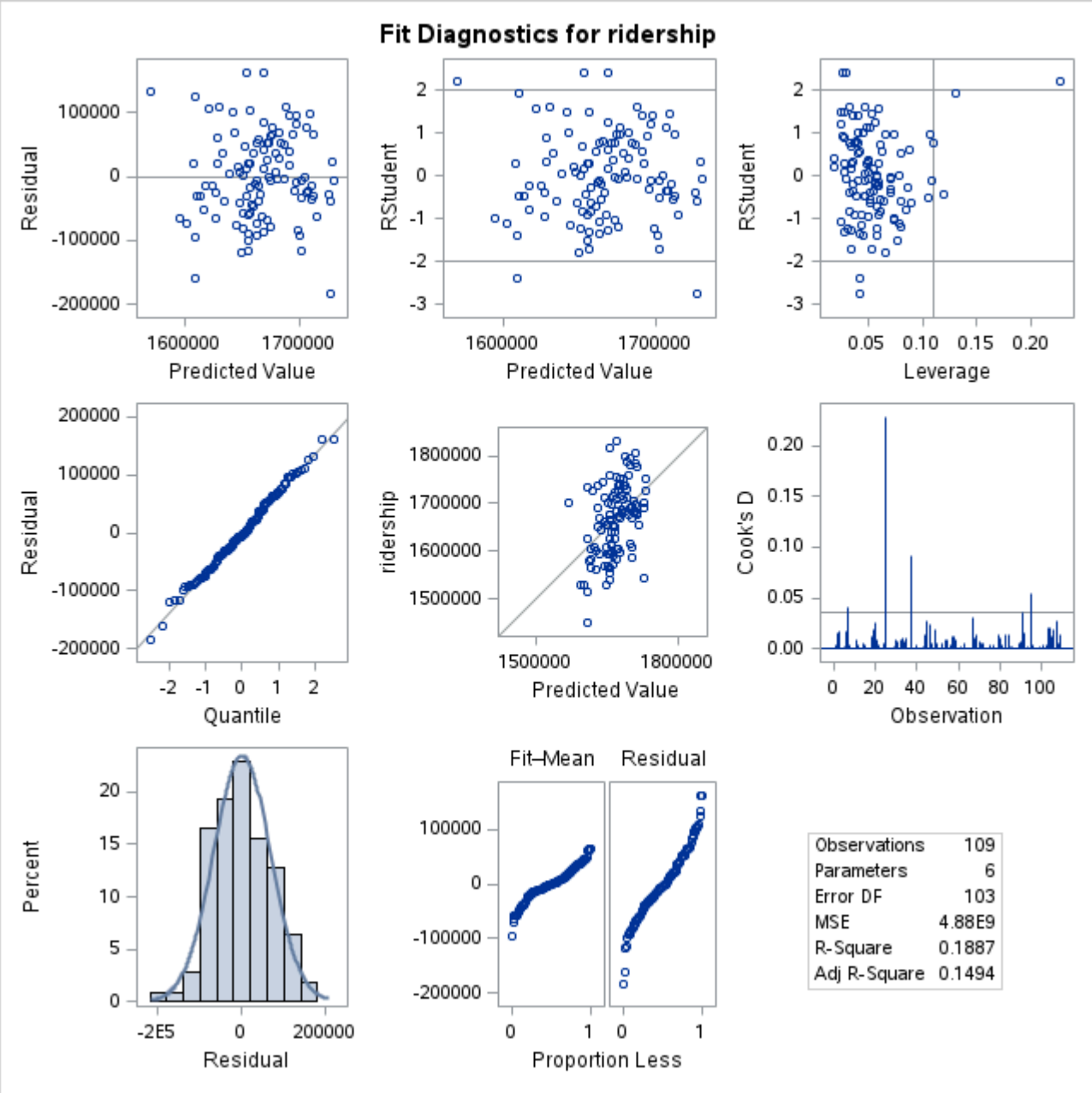
Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	1.168612E11	23372245288	4.79	0.0006
Error	103	5.022948E11	4876648287		
Corrected Total	108	6.19156E11			

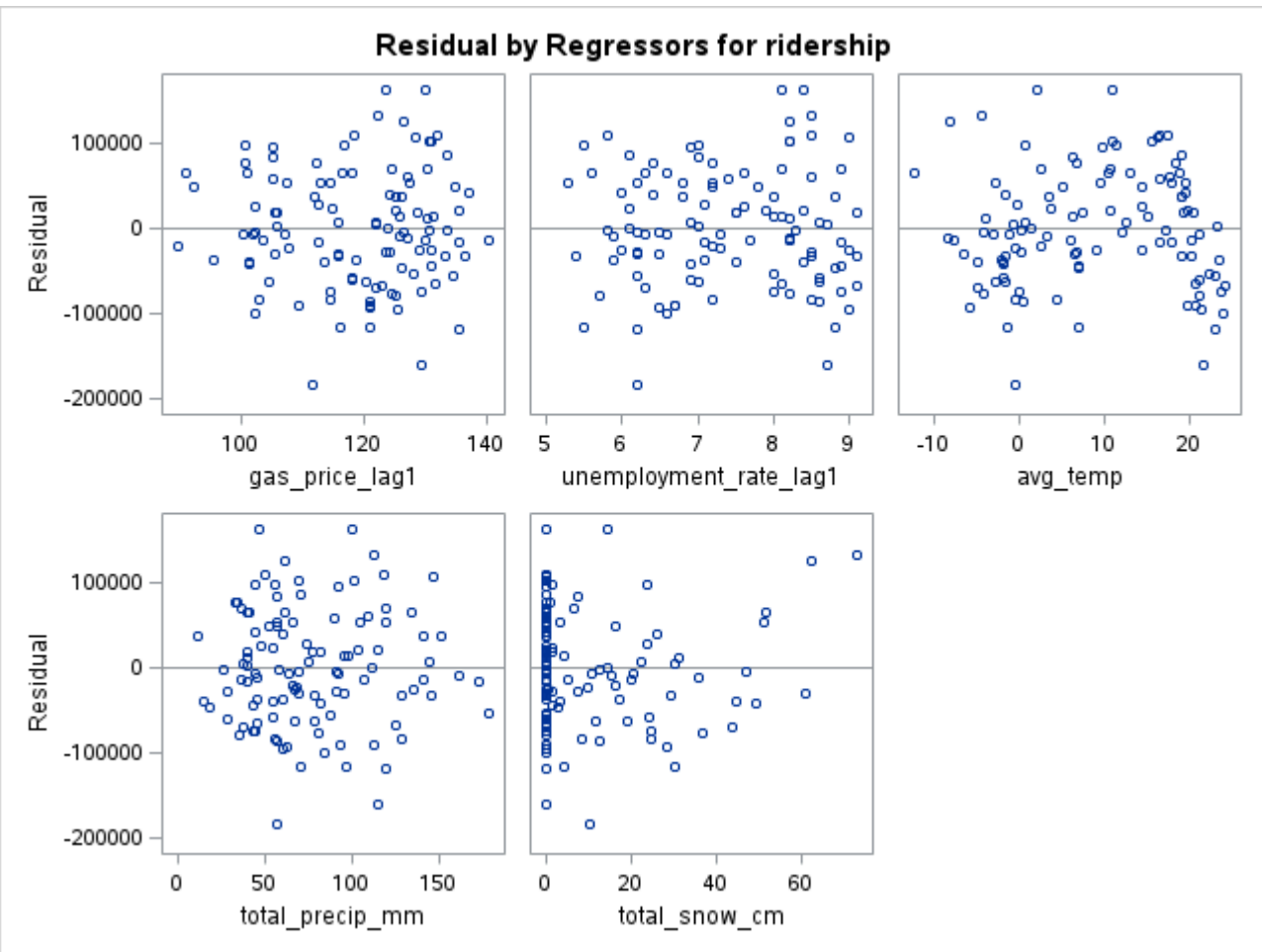
Root MSE	69833	R-Square	0.1887
Dependent Mean	1666000	Adj R-Sq	0.1494
Coeff Var	4.19166		

Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	1919828	75600	25.39	<.0001	.	0
gas_price_lag1	1	-472.45569	643.37155	-0.73	0.4644	0.77730	1.28650
unemployment_rate_lag1	1	-19164	6648.63562	-2.88	0.0048	0.88377	1.13152
avg_temp	1	-3537.31988	1144.77565	-3.09	0.0026	0.35011	2.85627
total_precip_mm	1	-58.67445	194.28718	-0.30	0.7633	0.87204	1.14674
total_snow_cm	1	-1911.73713	652.47152	-2.93	0.0042	0.39241	2.54833

Pre-Covid Model P4: Whithout Ridership History

The REG Procedure
Model: MODEL1
Dependent Variable: ridership


Fit Diagnostics for ridership
Residual
Predicted Value
RStudent
Leverage
Quantile
ridership
Cook's D
Observation
Percent
Fit–Mean
Residual
Proportion Less
Observations 109
Parameters 6
Error DF 103
MSE 4.88E9
R-Square 0.1887
Adj R-Square 0.1494


Residual by Regressors for ridership
Residual
gas_price_lag1
unemployment_rate_lag1
avg_temp
total_precip_mm
total_snow_cm